
Shunting Concentrates Path Gains and Improves Local Credit Assignment in Dendritic Networks

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Exact gradients for conductance-based dendritic trees factorize into a synapse-
2 local eligibility term and a single non-local compartment error; this factorization
3 is the foundation for local credit assignment in these networks. How well a so-
4 matic broadcast can substitute for the true compartment error depends on how
5 concentrated the conductance-stage path gains are across the tree. An oracle
6 experiment—replacing the broadcast with the exact conductance-stage transported
7 compartment errors—raises additive local learning from 33% to 98.5% on noise
8 resilience, nearly matching shunting (98.7%), strongly supporting path-gain fi-
9 delity as the dominant bottleneck rather than other architectural differences. Under
10 simple per-soma broadcast, shunting improves both gradient fidelity and end-task
11 learning relative to additive integration: cosine alignment rises from 0.006 to
12 0.202, and noise-resilience accuracy rises from 0.33 to 0.90. The advantage is
13 regime-dependent, growing with inhibitory conductance and largest under noisy
14 inputs. A structured pathway-vector extension (PV-LocalCA) further stabilizes
15 learned-routing performance on a hard cue-integration benchmark, reaching 99.1%
16 test accuracy. Together, these results identify conductance-stage path-gain concen-
17 tration via shunting inhibition as the key biophysical mechanism that determines
18 when low-bandwidth local credit assignment works in conductance-based dendritic
19 networks.

20 1 Introduction

21 Credit assignment in deep networks usually depends on backpropagation: global error transport
22 through symmetric pathways with no obvious biological substrate. Dendritic compartments suggest an
23 alternative. Each synapse has access to rich local state such as driving force, membrane conductance,
24 and branch voltage, while supervisory information could arrive as a low-bandwidth broadcast from
25 the soma [14]. The question here is when such local information is sufficient for useful learning.

26 Starting from conductance-based dendritic voltage equations [1], we derive exact gradients for
27 dendritic trees (Theorem 1) and show that each synaptic gradient factorizes into purely local eligibility
28 terms and a single non-local compartment error. Approximating that compartment error with a
29 broadcast signal yields a theorem-facing local rule (3F), together with practical heuristic wrappers
30 (4F/5F), and a structured pathway-vector extension for tasks where different branches must be credited
31 differently.

32 The central finding is mechanistic: shunting improves broadcast fidelity by concentrating conductance-
33 stage path gains. On MNIST, conductance-stage path-gain CV stays at 0.21–0.27 under shunting
34 versus 0.53–1.08 under additive integration, so one shared broadcast is a much better proxy for
35 compartment-specific errors under shunting. On the noise-resilience sweep, per-soma broadcast
36 achieves cosine alignment of about 0.20–0.32 under shunting, but stays near zero or negative under

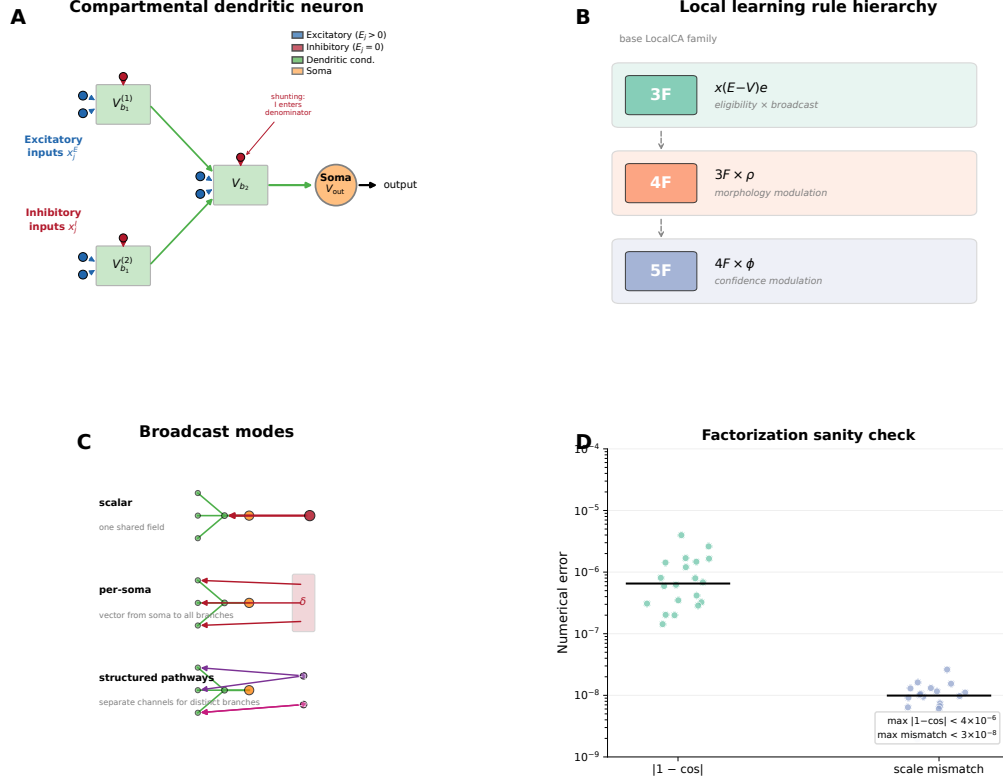


Figure 1: Model and credit assignment. (A) A compartmental dendritic neuron with excitatory (blue triangles) and inhibitory (red circles; shunting) synapses. Voltages propagate to the soma through learned dendritic conductances (green arrows). A somatic broadcast error e_n (dashed red) reaches all compartments; each synapse combines e_n with local quantities (presynaptic drive x_j , input resistance R_n^{tot} , and driving force $E_j - V_n$). (B) Compact LocalCA hierarchy: 3F is the theorem-facing local rule; 4F adds the heuristic morphology-aware factor ρ ; 5F further adds the heuristic confidence-like factor ϕ . (C) Broadcast-mode schematic. Scalar broadcast sends one shared field to all branches; per-soma broadcast sends a vector-valued somatic error to each branch; structured pathway feedback uses separate channels to preserve pathway identity when different branches should receive different teaching signals. (D) Headline exact-factorization summary. The panel states the factorization directly and reports the verification numbers compactly; Fig. 2D then shows the full diagnostic scatter across all theory-diagnostic conditions.

additive integration. Our question is therefore not whether local learning can match backpropagation on standard benchmarks, but what biophysical mechanism determines when low-bandwidth local credit assignment works in conductance-based dendritic networks. An oracle experiment that supplies the exact conductance-stage transported compartment errors strongly supports path-gain fidelity as the dominant bottleneck: additive local learning rises to 98.5% on noise resilience under the oracle, nearly matching shunting (98.7%).

We quantify these effects with a gradient-fidelity diagnostic based on directional alignment and scale mismatch between local and backprop gradients (Fig. 2). The absolute cosines remain modest, but they are consistently more positive under shunting and predict which local-learning regimes ultimately succeed. The paper’s contributions are: exact factorization for conductance-based dendritic trees, a mechanistic account of shunting as conductance-stage path-gain concentration, direct fidelity and oracle experiments that isolate the broadcast bottleneck, and a structured rank- K extension for settings where rank-1 broadcast fails.

2 Compartmental Voltage Model and Gradient Derivation

We use a steady-state conductance model derived from discretized passive cable dynamics [1, 2]. In normalized units, leak and inhibitory reversal are set to 0, the excitatory reversal is set to 1, and leak conductance is fixed at 1. Each compartment voltage is therefore a conductance-weighted average. Two properties of this form matter for what follows: local sensitivities depend on both driving force ($E - V$) and input resistance R^{tot} , and shunting inhibition corresponds to adding conductance when $E_{\text{inh}} \approx 0$.

2.1 Voltage Equation and Local Sensitivities

Consider compartment n with synaptic inputs j (activity x_j , reversal E_j , conductance $g_j^{\text{syn}} \geq 0$) and dendritic inputs from children (voltage V_j , conductance $g_j^{\text{den}} \geq 0$). The steady-state voltage is:

$$V_n = \frac{\sum_j E_j x_j g_j^{\text{syn}} + \sum_j V_j g_j^{\text{den}}}{\underbrace{\sum_j x_j g_j^{\text{syn}} + \sum_j g_j^{\text{den}} + 1}_{g_n^{\text{tot}}}}, \quad R_n^{\text{tot}} = 1/g_n^{\text{tot}}. \quad (1)$$

V_n is a convex combination of reversal potentials, child voltages, and leak. Writing $\mathcal{S}_n = \{0\} \cup \{E_j\}_j \cup \{V_j\}_j$, we have $\min \mathcal{S}_n \leq V_n \leq \max \mathcal{S}_n$ and $0 < R_n^{\text{tot}} \leq 1$. The local sensitivities follow directly:

Proposition 1 (Local Sensitivities).

$$\frac{\partial V_n}{\partial g_i^{\text{syn}}} = x_i R_n^{\text{tot}} (E_i - V_n), \quad \frac{\partial V_n}{\partial V_i} = g_i^{\text{den}} R_n^{\text{tot}}, \quad \frac{\partial V_n}{\partial g_i^{\text{den}}} = R_n^{\text{tot}} (V_i - V_n). \quad (2)$$

2.2 Shunting Inhibition as Divisive Gain Control

An inhibitory synapse with $E_{\text{inh}} \approx 0$ contributes current $(0 - V_n)x_j g_j^{\text{syn}}$ and increases g_n^{tot} . Its sensitivity is $\partial V_n / \partial g_j^{\text{syn}} = -x_j R_n^{\text{tot}} V_n$, which corresponds to multiplicative attenuation (divisive normalization). Shunting is divisive at the voltage level, but its effect on firing rates can be subtractive in some regimes [7]. Inhibitory plasticity can balance excitation dynamically [8]; our learned inhibitory conductances serve an analogous role.

2.3 Exact Gradients for Dendritic Trees

Let V_0 be the somatic output, $\hat{y} = W_{\text{dec}} V_0$ the decoder, and $\delta_0 = W_{\text{dec}}^\top (\partial L / \partial \hat{y})$ the somatic error.

Theorem 1 (Conductance-Stage Backpropagation on a Dendritic Tree). *For a rooted dendritic tree with soma at node 0 and unique parent $p(n)$ for each non-somatic compartment, if each branch passes its pre-reactivation voltage directly to its parent, then the loss gradient satisfies*

$$\frac{\partial L}{\partial V_n} = \frac{\partial L}{\partial V_{p(n)}} R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}, \quad (3)$$

with boundary condition $\partial L / \partial V_0 = \delta_0$. Because the graph is a tree, each compartment has a unique path to the soma. Defining the path gain

$$\alpha_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightsquigarrow 0)} R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad \frac{\partial L}{\partial V_n} = \alpha_n \delta_0, \quad (4)$$

with $\alpha_0 = 1$.

Proof. Apply the chain rule on the tree-structured computation graph using Prop. 1. $\square$

Corollary 1 (Local-Global Factorization). *The exact synaptic gradient at compartment n factorizes as:*

$$\frac{\partial L}{\partial g_i^{\text{syn}}} = \underbrace{x_i R_n^{\text{tot}} (E_i - V_n)}_{\text{synapse-local eligibility}} \cdot \underbrace{\frac{\partial L}{\partial V_n}}_{\text{compartment error}}, \quad (5)$$

where the eligibility term depends only on quantities available at synapse i (presynaptic activity x_i , input resistance R_n^{tot} , driving force $E_i - V_n$), and the compartment error $\partial L / \partial V_n$ is the sole non-local quantity.

Under identity upward transfer, this factorization isolates the only non-local quantity as $\partial L / \partial V_n = \alpha_n \delta_0$. With a post-voltage activation $a_n = f_n(V_n)$ transmitted to the parent, the exact recursion becomes $\partial L / \partial V_n = f'_n(V_n) (\partial L / \partial V_{p(n)}) R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}$, so the compartment error remains the sole non-local term but the effective path gain acquires local activation-derivative factors. Any practical local rule therefore depends entirely on how well a broadcast signal approximates that exact compartment error, which we evaluate in Sec. 4.2. In the models studied here, shunting improves this approximation by reducing and stabilizing R_n^{tot} , which narrows the spread of local sensitivities across the tree. The theorem is stated for the pre-reactivation compartment voltage V_n produced by the conductance stage itself. In the empirical networks, each dendritic branch layer may optionally apply a post-voltage reactivation $f(V_n)$ before passing activity to the next stage; unless otherwise noted, the manuscript sweeps use a learnable bounded `param_tanh` reactivation. Our exact-error diagnostics are computed on the hooked pre-reactivation voltage, so the factorization check and compartment-error analyses target the conductance-stage quantity defined above rather than the post-reactivation activity.

3 Local Learning Rules

3.1 Broadcast Error Approximation

We approximate the exact compartment error $\partial L / \partial V_n$ (Corollary 1) with a broadcast signal e_n derived from the somatic error $\delta_0 = W_{\text{dec}}^\top (\partial L / \partial \hat{y})$. We consider five broadcast modes of increasing structure:

- **Scalar:** $e_n = \bar{\delta} \cdot \mathbf{1}$, where $\bar{\delta} = \text{mean}(\delta_0)$ is a single scalar broadcast to all compartments.
- **Per-soma:** $e_n = \delta_0$ when the layer dimension matches the output; layers with mismatched dimensions fall back to scalar.
- **Low-rank:** $e_n = \Gamma_K(\delta_0)$, where Γ_K uses a small number K of random broadcast channels to project the somatic error into a layer-specific vector field.
- **Local mismatch:** $e_n = (1 - \alpha) \bar{\delta} \cdot \tilde{m}_n + \alpha \bar{\delta}$, where $\tilde{m}_n$ is the RMS-normalized, batch-centered parent-child voltage difference $P_n - V_n$ and $\alpha = 0.2$.
- **Pathway-vector:** $e_n = \alpha \bar{\delta} \cdot \mathbf{1} + (1 - \alpha) \Gamma_n(\delta_0)$, where Γ_n projects the somatic error into router-inferred pathway roles, gates those roles by local pathway activity, and transports the resulting vector hierarchically through the block-structured dendritic tree.

We use per-soma broadcast by default. In our experiments, the local-mismatch variant performs much worse (Appendix A), indicating that broadcast quality is critical. Under shunting, this simple broadcast is often sufficient on standard classification tasks, but the cue-routing benchmark in Sec. 4.7 requires the structured pathway-vector form. Appendix Table S10 summarizes the broader set of implemented model families, training strategies, and broadcast modes used throughout the study.

3.2 Three-Factor Rule (3F)

Definition 1 (3F Update). For synaptic and dendritic conductances:

$$\Delta g_j^{\text{syn}} = \eta \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B, \quad \Delta g_j^{\text{den}} = \eta \langle R_n^{\text{tot}} (V_j - V_n) e_n \rangle_B, \quad (6)$$

where $\langle \cdot \rangle_B$ denotes the batch average.

The three factors are presynaptic activity x_j (or a voltage difference), postsynaptic modulation through driving force and input resistance, and the broadcast error e_n . The same rule applies to excitatory and inhibitory synapses. The sign difference comes from the driving force $(E_j - V_n)$.

Additive control. Rule (6) is the local gradient of the *shunting* voltage $V_n = \sum E_j g_j x_j / g_n^{\text{tot}}$. For the additive control ($V_n = \sum g_j x_j$, no divisive normalization), the correct local gradient is simpler:

125 $\Delta g_j^{\text{syn}} = \eta \langle x_j e_n \rangle_B$, with no driving-force or R_n^{tot} terms ($R_n^{\text{tot}} \equiv 1$ by definition since there is no
 126 denominator). Throughout, each architecture uses the learning rule derived from its own forward-pass
 127 dynamics. This keeps the comparison architecture-matched rather than applying a shunting-derived
 128 rule to an additive model.

129 3.3 Practical Heuristic Wrappers: 4F and 5F

130 **4F (heuristic attenuation proxy).** In the conductance-stage recursion (4), the path gain α_n attenuates
 131 the somatic error differently at each compartment. To compensate without explicitly computing that
 132 gain, 4F uses an online correlation proxy

$$\rho_n = \text{Cov}(\bar{V}_n, \bar{V}_0) / (\sqrt{\text{Var}(\bar{V}_n) \text{Var}(\bar{V}_0)} + \varepsilon).$$

133 High ρ_n up-weights branches whose voltages covary with the soma; low ρ_n down-weights branches
 134 where the broadcast is likely to be a poor proxy.

135 **5F (confidence-weighted 4F).** 5F adds a second heuristic factor

$$\phi_n = \text{Var}(V_n) / (\sigma_{\text{res}}^2 + \varepsilon),$$

136 where σ_{res}^2 is the residual variance of V_n after linear regression on the parent voltage. Operationally,
 137 ϕ_n is a branch-wise confidence gate: it becomes large when local voltages are strong and stable
 138 relative to unexplained residual fluctuations, and small when local fluctuations dominate. We clamp it
 139 to $[0.25, 4.0]$ for stability. This factor is empirical, not theorem-derived. The paper’s core claim rests
 140 on path-gain concentration and the oracle experiment; the gain from ϕ_n should be read as practical
 141 denoising rather than theoretical necessity.

Proposition 2 (5F Update).

$$\Delta g_j^{\text{syn}} = \eta \rho_n \phi_n \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B. \quad (7)$$

142 A feedback-alignment-style random-broadcast argument is included in Appendix C; it is supportive
 143 but not central to the main mechanism claim.

144 3.4 Structured pathway feedback and branch-role plasticity

145 For tasks with latent pathway structure, scalar broadcast can erase the distinction between different
 146 dendritic branches that should be credited differently. Recent experimental evidence supports the
 147 existence of vectorized instructive signals in cortical dendrites [43], motivating the structured feedback
 148 variant we introduce here. We therefore augment the local rule with a structured cue-routing variant,
 149 *PV-LocalCA*.

150 **Hierarchical pathway-vector broadcast.** The encoder/router defines a low-dimensional role basis
 151 over latent pathways. Instead of collapsing δ_0 to a scalar or per-soma vector, we project it into this
 152 role basis and gate each role by local pathway activity:

$$e_n^{\text{PV}} = \sum_{k=1}^K q_{n,k} c_k,$$

153 where $c_k = \langle r_k, \delta_0 \rangle$ are role coefficients and $q_{n,k}$ are local branch-role weights. Scalar broadcast is
 154 the special case $K=1$, while PV-LocalCA is a structured rank- K extension rather than a separate
 155 learning principle.

156 **Router-derived branch roles.** Rather than assigning hand-coded apical/basal labels, each branch
 157 j receives a normalized role profile r_j inferred from router assignments and incoming block weights.
 158 Plasticity is then gated by branch selectivity and by a synapse-specific role-alignment factor $\langle r_i, r_j \rangle$,
 159 biasing learning toward pathway-consistent afferents without imposing a fixed branch taxonomy.

160 4 Experiments

161 4.1 Setup

162 We evaluate two settings: a *capacity-calibrated* regime, where matched architectures achieve strong
 163 backprop performance, and *controlled sweeps* that isolate inhibition strength, broadcast mode, and

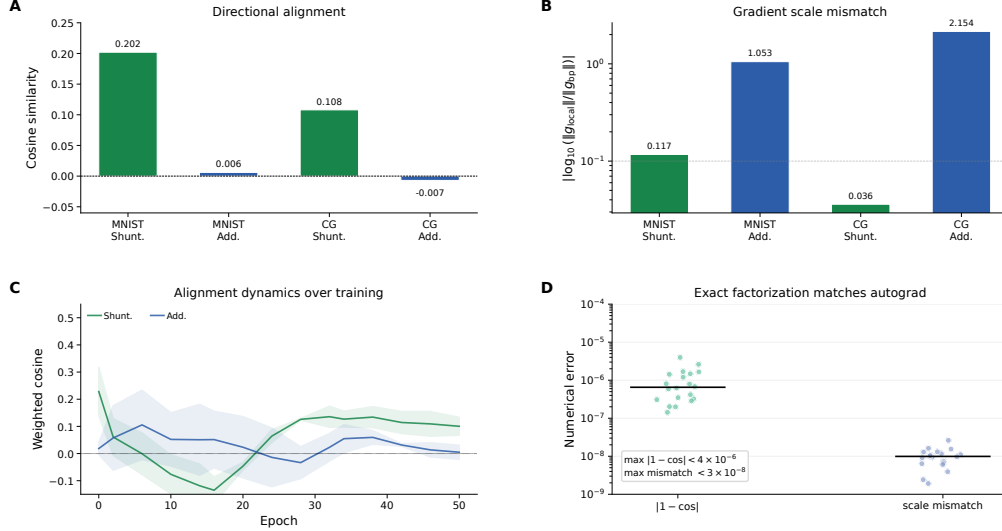


Figure 2: Gradient-fidelity diagnostic. (A) Weighted cosine similarity between local and backprop gradients: shunting (green) is consistently more aligned than additive (blue) on both MNIST and context gating. (B) Gradient scale mismatch on the same matched-weight comparisons. Shunting is not only better aligned, but also substantially closer in norm. (C) Per-layer cosine similarity over training epochs. Shunting proximal layers approach high positive alignment; additive layers remain near zero. (D) Exact-factorization sanity check across all theory-diagnostic conditions. Reconstructed gradients match autograd to numerical precision, showing that LocalCA’s only approximation is in the broadcast substitute for the compartment error.

164 architecture. Our main datasets are MNIST, Fashion-MNIST [37], and four synthetic tasks: *context*
 165 *gating*, *noise resilience*, *info shunting*, and *cue integration*; formal definitions are in Appendix I.
 166 Architectures include point MLP baselines and dendritic cores with either additive integration or
 167 shunting. Main comparisons are architecture-matched additive vs. shunting cores and local vs.
 168 backprop training; DFA appears only as a supplementary baseline. Most headline MNIST, Fashion-
 169 MNIST, and context-gating results use 5F with per-soma broadcast. Unless otherwise stated, dendritic
 170 layers also use the learnable post-voltage reactivation `param_tanh`; Appendix Tables S7–S8 then
 171 audit activation shape in a more controlled setting by comparing identity (none) against a frozen
 172 `param_tanh` under matched additive/shunting initialization. The cue-routing benchmark instead
 173 uses PV-LocalCA, and the inhibition sweeps additionally use a transported-error oracle derived from
 174 Theorem 1 as an upper bound rather than as the proposed biological rule. Headline mechanistic and
 175 routed results use 3–5 seeds; 1–2 seed screens are treated as exploratory and are either omitted from
 176 the main claims or explicitly labeled as such in the appendix. The experiments follow one chain:
 177 first verify how closely different broadcasts track the exact gradients implied by the theorem, then
 178 test whether those fidelity differences translate into learning, and finally study a structured-feedback
 179 extension for settings where rank-1 broadcast fails.

180 4.2 From Exact Factorization to Credit-Signal Fidelity

181 To test whether these performance differences correspond to better credit signals, we compare
 182 local and backprop gradients on the same batch at the same parameter values. For each parameter
 183 tensor p , we compute directional alignment (cosine similarity) and scale mismatch, defined as
 184 $|\log_{10}(\|g_p^{\text{local}}\|/\|g_p^{\text{bp}}\|)|$, aggregated by parameter count.

185 Across matched-weight comparisons, shunting is consistently more aligned with backprop and
 186 substantially closer in norm than the additive control: on MNIST, weighted cosine rises from 0.006
 187 to 0.202 and scale mismatch falls from 1.053 to 0.117; on context gating, cosine rises from -0.007
 188 to 0.108 and scale mismatch falls from 2.154 to 0.036. Figure 2C shows that this advantage appears
 189 throughout training, especially in proximal layers and dendritic conductances. Figure 2D confirms that
 190 the theory is operational in code: reconstructing gradients from Corollary 1 using exact compartment
 191 errors matches autograd to numerical precision (weighted cosine 1.000001 ± 0.000001 , weighted

scale mismatch $< 4 \times 10^{-8}$). The only approximation made by LocalCA is therefore the broadcast substitute for $\partial L / \partial V_n$.

4.3 Mechanistic Chain: Path Gains, Broadcast Fidelity, and Oracle Transport

Figure 3 provides the paper’s central empirical chain: shunting concentrates conductance-stage path gains, which improves direct compartment-error fidelity, which in turn makes low-bandwidth local learning effective; the transported-error oracle then isolates how much of the remaining gap is due to broadcast quality rather than other architectural differences.

Panel A measures the coefficient of variation of the exact conductance-stage path gains α_n across compartments. On MNIST, shunting keeps this distribution tightly concentrated across inhibition levels (CV about 0.21–0.27), whereas the additive control remains much broader (CV about 0.53–1.08). Panel B then measures the sole non-local term from Corollary 1, $\partial L / \partial V_n$, directly. On the noise-resilience sweep, per-soma broadcast tracks the exact compartment error much better under shunting than under additive integration, rising from about 0.20 at $N_I=0$ to about 0.32 near the best-performing inhibition range, while additive stays near zero or negative. The transported-error oracle obtained by recursively applying those conductance-stage path gains improves fidelity further in both architectures, showing that the theorem defines an operational upper bound on low-bandwidth broadcast.

Panel C trains the same local rule using that oracle signal. On noise resilience, additive learning rises from 33–47% under per-soma broadcast to 75.6% at $N_I=0$ and 98.5% at $N_I=40$ under transported broadcast. Shunting improves from 71.7–90.3% under per-soma to 98.5–99.3% across the same sweep. On MNIST, transported broadcast reaches 94.6% for additive and 96.8% for shunting. These oracle results strongly support path-gain fidelity as the dominant bottleneck for LocalCA performance, while the smaller remaining additive–shunting gap under matched transported broadcast suggests an additional contribution from conductance-shaped local sensitivities.

Panel D shows that coarse broadcast can still be enough once local sensitivities are well scaled: 4-bit quantization preserves full performance (61.6% vs. 61.7%; $p > 0.9$), while sign-only (37.5%) and sparse top-30% (38.2%) degrade sharply. Normalization-only additive controls and DFA baselines (Appendix Figs. S9, S7) improve over weaker baselines without matching the shunting conductance model.

4.4 Capacity-Calibrated Competence

Having established the mechanism, we next ask whether the same conditions improve end-task learning in architectures that are already competent under backpropagation. In the capacity-calibrated regime, shunting dendritic cores remain reasonably competitive under local learning: on MNIST, Fashion-MNIST, and context gating, the best local configuration (5F, per-soma broadcast, local decoder) stays within about 5–7 percentage points of the matched backprop ceiling (Appendix Table S4; Appendix Fig. S2). Within the local-rule family, 5F consistently outperforms 4F and 3F (Appendix Table S1), and per-soma strongly outperforms scalar and local-mismatch broadcast (Appendix Table S3). We therefore treat 5F as the strongest practical wrapper in this paper, not as a uniquely derived optimum; its clamp bounds and residual estimator remain heuristic design choices, although the main tuning knobs are not especially fragile (Appendix Fig. S4). Appendix Table S2 summarizes the empirical division of labor across architecture, rule, and broadcast choices.

4.5 Regime Dependence of the Shunting Advantage

The benefit of shunting varies across tasks. On MNIST with matched per-soma broadcast, shunting outperforms additive by about 2 percentage points; a similar pattern holds on Fashion-MNIST (81.1% vs. 79.4%). On tasks that require credit signals robust to noise, the gap is much larger: +50.3 pp on noise resilience (with $N_I=10$ inhibitory synapses per branch) and +24.8 pp on info shunting ($N_I=0$). Appendix Fig. S2 shows the full dose-response: the shunting advantage grows with inhibitory conductance strength on the inhibition-sensitive tasks. Additive cores can still perform reasonably on easier settings, reaching about 89% on MNIST after tuning, but they fail in regimes where inhibition-mediated normalization matters for gradient propagation.

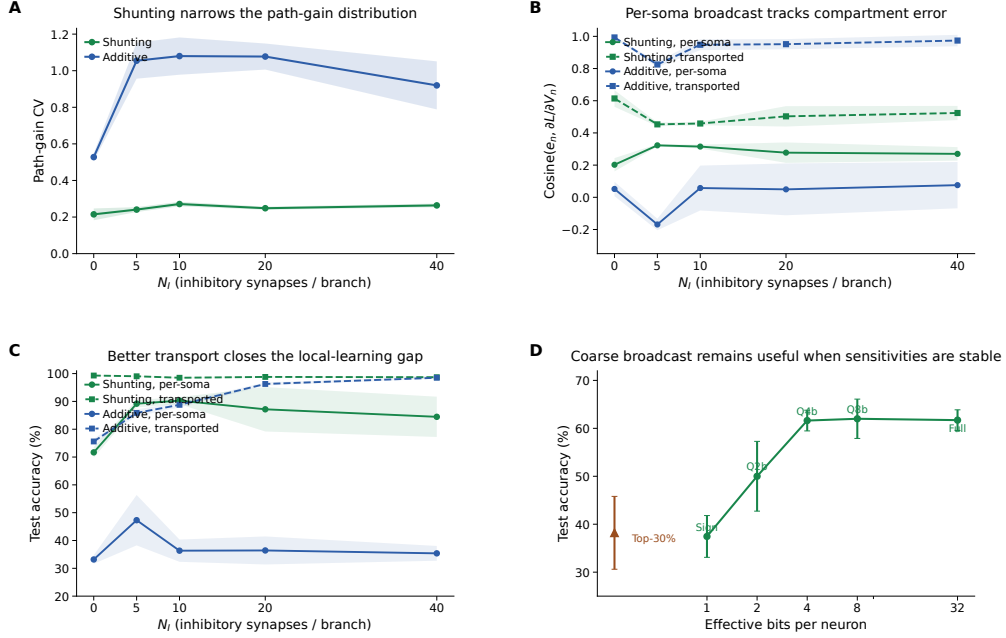


Figure 3: **Mechanistic evidence.** (A) Conductance-stage path-gain dispersion on MNIST, measured as the coefficient of variation of the exact tree attenuation α_n across compartments. Shunting keeps these conductance-stage path gains much more concentrated than the additive control as inhibition increases. (B) Direct compartment-error fidelity on noise resilience. Solid lines compare the per-soma broadcast e_n to the exact compartment error $\partial L / \partial V_n$; dashed lines show the transported-error oracle obtained by recursively applying conductance-stage path gains. Shunting substantially improves the fidelity of the simple per-soma broadcast, while the transported oracle defines the remaining approximation gap. (C) Learning with the transported-error oracle on noise resilience. Improving the broadcast field is enough to recover near-ceiling local learning, especially in the additive architecture where per-soma broadcast fails most severely. (D) Broadcast bandwidth. Four-bit quantization preserves full performance; only sign-only and sparse modes degrade sharply.

242 4.6 Additional Stress Tests

243 Across three additional stress tests — increased dendritic depth, noisy broadcast signals, and Fashion-
 244 MNIST — the same qualitative pattern persists: shunting degrades more gracefully than additive,
 245 while the largest gains still appear on tasks that stress the credit signal most strongly. We keep the
 246 full curves in Appendix Fig. S6 because they support, rather than define, the core mechanistic claim.

247 4.7 Extension: Structured Feedback When Per-Soma Broadcast Fails

248 The main paper focuses on low-bandwidth scalar or per-soma broadcast. We now consider a harder
 249 cue-integration benchmark where that approximation is insufficient because different dendritic
 250 pathways must be credited differently. The task asks the model to combine two noisy sensory cues
 251 using a context signal that indicates which cue is reliable on each trial. In the fixed-pathway control,
 252 context is copied into both cue pathways. In the learned-pathway setting, the router must instead
 253 discover that separation and align one latent pathway with each cue stream.

254 Figure 4 summarizes five-seed runs for this benchmark. The benchmark itself is learnable: fixed-
 255 pathway additive local learning remains near ceiling ($99.1 \pm 0.3\%$), learned-routing additive local
 256 learning also succeeds ($98.1 \pm 2.4\%$), and standard learned-router backprop training stays near
 257 $99.2 \pm 0.4\%$ for both additive and shunting cores. The specific failure appears when learned-routing
 258 shunting LocalCA uses the default per-soma broadcast. At the original 90-epoch setting, accuracy is
 259 only $82.6 \pm 14.2\%$, and modest router or optimization changes help only slightly: low temperature
 260 gives $87.6 \pm 13.3\%$, freeze-20 gives $83.8 \pm 13.9\%$, and a tuned per-soma run reaches $88.2 \pm 13.9\%$.

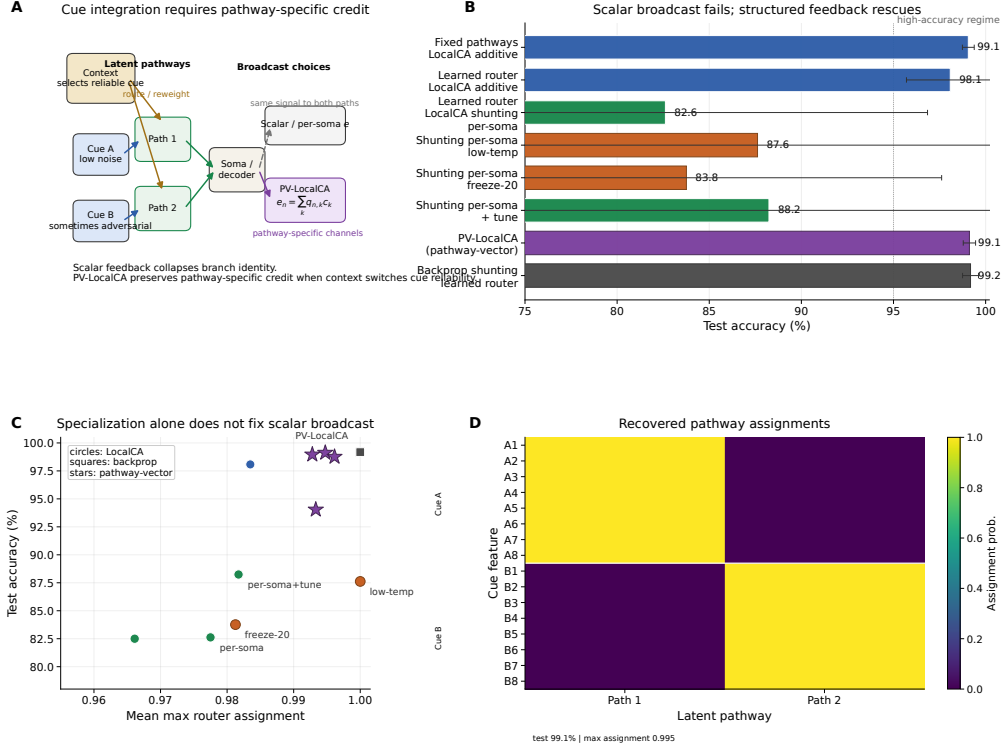


Figure 4: Structured feedback for pathway discovery on cue integration. (A) Task schematic. Context determines which noisy cue is reliable on each trial. Scalar or per-soma broadcast sends the same teaching signal to both pathways, whereas PV-LocalCA preserves pathway identity through a low-rank pathway-vector feedback field. (B) Seed-averaged diagnosis and focused tuning (5 seeds per condition). Learned-routing additive local learning remains high, but learned-routing shunting with per-soma broadcast stays unstable; low-temperature, freeze-20, and per-soma tuning help only modestly, whereas tuned pathway-vector feedback reaches $99.1 \pm 0.3\%$. (C) Test accuracy vs. router specialization across learned-router conditions. High specialization alone does not guarantee success: per-soma, low-temperature, and freeze-20 runs remain inaccurate despite near-discrete assignments. (D) Recovered pathway assignments for a representative tuned pathway-vector run. Cue-A and cue-B features segregate cleanly into separate latent pathways.

Replacing per-soma broadcast with PV-LocalCA changes the regime. The untuned 90-epoch version improves to $94.0 \pm 11.0\%$, and the tuned setting (140 epochs, learning rate 10^{-3}) reaches $99.1 \pm 0.3\%$ test and $99.4 \pm 0.2\%$ validation accuracy while maintaining strong specialization (mean max assignment 0.995 ± 0.008). The failure is therefore not pathway discovery itself; it is the loss of pathway identity under rank-1 feedback. Consistent with that interpretation, simply increasing the number of random low-rank channels does not recover the same gain (Appendix Table S6). When the feedback preserves that structure, learned-routing shunting LocalCA becomes reliable.

5 Related Work

High-performance local learning. Recent methods such as AugLocal, CCL, and DLL have substantially raised the benchmark-performance bar for local learning on standard architectures [33, 34, 36]. Our aim is therefore not to claim the strongest benchmark accuracy among local-learning methods, but to identify the biophysical mechanism that determines when low-bandwidth credit signals become useful in conductance-based dendritic networks.

Local credit-assignment dynamics. Random feedback, DFA, DTP, DFC, equilibrium-style methods, and predictive-coding approaches solve locality through different feedback or inference dynamics [9, 10, 13, 15, 16, 25–29, 32]. EBD [35] is especially close in spirit because it studies when di-

rect error broadcast can be made useful by shaping the broadcast field itself. Our contribution is complementary: rather than changing the broadcast objective or inference dynamics, we identify a conductance-based inductive bias—conductance-stage path-gain concentration under shunting—that makes even a simple low-bandwidth broadcast more faithful to the exact compartment error.

Dendrites for robustness and context. Dendritic trees support nonlinear computation [3, 4, 30] and have inspired learning schemes based on segregated dendrites, dendritic prediction errors, burst-dependent plasticity, and latent equilibrium [5, 11, 12, 22–24]. Outside supervised iid classification, active-dendrite architectures can also improve robustness in dynamic multi-task settings where point-MLP baselines suffer interference [44]. Our cue-routing extension asks a related question—when dendritic structure helps preserve pathway-specific computation—but uses conductance-based normalization and structured credit signals rather than sparse kWTA gating.

Neuroscience grounding. Shunting implements divisive normalization [6], though its rate effects can be subtractive in some regimes [7]; inhibitory conductance also modulates gain and signal-to-noise [31]. Recent evidence for vectorized instructive signals in cortical dendrites [43] further supports the structured-feedback extension studied here. To our knowledge, prior work has not tied exact conductance-stage path gains, direct gradient-fidelity diagnostics, and shunting inhibition into one causal account of when local dendritic learning works.

6 Discussion

We started from exact gradients for a conductance-based dendritic tree and asked when a somatic broadcast can serve as a useful proxy for compartment-specific errors. In the models studied here, the answer depends strongly on the inhibitory regime. Shunting changes the conductance denominator, stabilizes effective local sensitivities, improves agreement between local and backprop gradients, and yields better local learning than an additive control, especially in noisy or inhibition-dependent settings. An exploratory morphology $\times$ inhibition sweep (Appendix Fig. S10) suggests that tree shape shifts this operating regime: shallow trees peak at lower inhibition, while deeper trees retain their largest shunting advantage at modest inhibition. Explicit normalization recovers part of this benefit but not all of it, suggesting that shunting changes more than overall scale alone. This is therefore a mechanism paper, not a benchmark paper: the main contribution is to explain when simple broadcast-based local learning works in conductance dendritic networks. The appendix activation audit further shows that this mechanism should be interpreted in the bounded `param_tanh` operating regime used throughout the main experiments: even when that activation is frozen rather than learned, changing the post-voltage nonlinearity can materially alter both learning and credit-signal fidelity.

Limitations. Our best accuracy (91.4% MNIST, 81.1% Fashion-MNIST) remains below several recent local-learning methods that operate in abstract compartments or standard activation spaces [33–36]. This reflects the constraints of conductance-based voltage space: bounded voltages, positive conductances, and a denominator-heavy computation graph. In the conservative flattened-CIFAR-10 sweep in Appendix Fig. S8, shunting local learning reaches 21.2% versus 40.1% backprop, indicating that the gradient-fidelity mechanism alone is insufficient for harder tasks without architectural advances such as convolutional encoders. Naive local broadcasts remain much weaker than per-soma on MNIST, and the cue-routing benchmark requires a still more structured pathway-vector signal (Sec. 4.7); our claims are therefore specific to the broadcast family studied here, not arbitrary local mismatch heuristics. Per-soma broadcast is low-bandwidth but still neuron-global because it assumes access to the full somatic error vector. Depth remains challenging: local learning degrades more than backprop as trees deepen (Appendix Fig. S6A), and at depth 4 shunting local learning is still roughly 33 percentage points below backprop. Shunting networks are also *not* gain-invariant in practice: exploratory global gain-shift pilots show sharp degradation because leak conductance breaks exact gain invariance. Finally, standard FA is incompatible with dendritic layer geometry (Appendix Fig. S7), and the weight-statistics analysis in Appendix G is descriptive rather than central evidence.

Broader relevance. The results are relevant to neuroscience, machine learning, and neuromorphic hardware. For neuroscience, they suggest a connection between divisive normalization and local credit fidelity. For machine learning, they show that conductance-based inductive biases can shape

328 gradient geometry in ways that benefit local learning. For neuromorphic implementations, the update
329 rules remain strictly local and parallelizable.

330 **Testable predictions.** Our framework predicts that (1) stronger perisomatic inhibition yields more
331 precise synaptic plasticity; (2) GABA_A blockade selectively impairs multi-layer credit tasks; (3) the
332 dendritic–somatic voltage correlation (ρ_n) increases during learning; and (4) learned excitatory
333 weight distributions follow log-normal statistics whose width depends systematically on inhibitory
334 conductance (Appendix G).

335 References

- 336 [1] Koch, C. (1999). *Biophysics of Computation*. Oxford University Press.
- 337 [2] Dayan, P., & Abbott, L. F. (2001). *Theoretical Neuroscience*. MIT Press.
- 338 [3] Poirazi, P., Brannon, T., & Mel, B. W. (2003). Pyramidal neuron as two-layer neural network.
339 *Neuron*, 37(6), 989–999.
- 340 [4] London, M., & Häusser, M. (2005). Dendritic computation. *Annual Review of Neuroscience*, 28,
341 503–532.
- 342 [5] Urbanczik, R., & Senn, W. (2014). Learning by the dendritic prediction of somatic spiking.
343 *Neuron*, 81(3), 521–528.
- 344 [6] Carandini, M., & Heeger, D. J. (2012). Normalization as a canonical neural computation. *Nature*
345 *Reviews Neuroscience*, 13(1), 51–62.
- 346 [7] Holt, G. R., & Koch, C. (1997). Shunting inhibition does not have a divisive effect on firing
347 rates. *Neural Computation*, 9(5), 1001–1013.
- 348 [8] Vogels, T. P., Sprekeler, H., Zenke, F., Clopath, C., & Gerstner, W. (2011). Inhibitory plas-
349 ticity balances excitation and inhibition in sensory pathways and memory networks. *Science*,
350 334(6062), 1569–1573.
- 351 [9] Lillicrap, T. P., Cownden, D., Tweed, D. B., & Akerman, C. J. (2016). Random synaptic
352 feedback weights support error backpropagation for deep learning. *Nature Communications*, 7,
353 13276.
- 354 [10] Nøkland, A. (2016). Direct feedback alignment provides learning in deep neural networks.
355 *NeurIPS*, 29.
- 356 [11] Guerguiev, J., Lillicrap, T. P., & Richards, B. A. (2017). Towards deep learning with segregated
357 dendrites. *eLife*, 6, e22901.
- 358 [12] Sacramento, J., et al. (2018). Dendritic cortical microcircuits approximate the backpropagation
359 algorithm. *NeurIPS*, 31.
- 360 [13] Whittington, J. C., & Bogacz, R. (2019). Theories of error back-propagation in the brain. *Trends*
361 *in Cognitive Sciences*, 23(3), 235–250.
- 362 [14] Richards, B. A., & Lillicrap, T. P. (2019). Dendritic solutions to the credit assignment problem.
363 *Current Opinion in Neurobiology*, 54, 28–36.
- 364 [15] Scellier, B., & Bengio, Y. (2017). Equilibrium propagation. *Frontiers in Computational Neuro-*
365 *science*, 11, 24.
- 366 [16] Song, Y., et al. (2024). Inferring neural activity before plasticity as a foundation for learning
367 beyond backpropagation. *Nature Neuroscience*, 27, 348–358.
- 368 [17] Gretton, A., et al. (2005). Measuring statistical dependence with Hilbert-Schmidt norms. *ALT*,
369 63–77.
- 370 [18] Frémaux, N., & Gerstner, W. (2016). Neuromodulated spike-timing-dependent plasticity, and
371 theory of three-factor learning rules. *Frontiers in Neural Circuits*, 9, 85.

- [19] Bellec, G., et al. (2020). A solution to the learning dilemma for recurrent networks of spiking neurons. *Nature Communications*, 11, 3625.
- [20] Welford, B. P. (1962). Note on a method for calculating corrected sums of squares and products. *Technometrics*, 4(3), 419–420.
- [21] Turrigiano, G. G. (2008). The self-tuning neuron: synaptic scaling of excitatory synapses. *Cell*, 135(3), 422–435.
- [22] Payeur, A., et al. (2021). Burst-dependent synaptic plasticity can coordinate learning in hierarchical circuits. *Nature Neuroscience*, 24(7), 1010–1019.
- [23] Greedy, W., et al. (2022). Single-phase deep learning in cortico-cortical networks. *NeurIPS*, 35.
- [24] Haider, P., et al. (2021). Latent equilibrium. *NeurIPS*, 34.
- [25] Hinton, G. (2022). The forward-forward algorithm. *arXiv:2212.13345*.
- [26] Dellaferrera, G., & Kreiman, G. (2022). Error-driven input modulation: Solving the credit assignment problem without a backward pass. *Proceedings of the 39th International Conference on Machine Learning, Proceedings of Machine Learning Research*, 162, 4937–4955.
- [27] Lee, D.-H., et al. (2015). Difference target propagation. *ECML*, 498–515.
- [28] Meulemans, A., et al. (2021). Credit assignment in neural networks through deep feedback control. *NeurIPS*, 34.
- [29] Millidge, B., Seth, A. K., & Buckley, C. L. (2021). Predictive coding: A theoretical and experimental review. *arXiv:2107.12979*.
- [30] Koch, C., Poggio, T., & Torre, V. (1983). Nonlinear interactions in a dendritic tree. *PNAS*, 80(9), 2799–2802.
- [31] Silver, R. A. (2010). Neuronal arithmetic. *Nature Reviews Neuroscience*, 11(7), 474–489.
- [32] Max, K., Kriener, L., Pineda García, G., Nowotny, T., Jaras, I., Senn, W., & Petrovici, M. A. (2024). Learning efficient backprojections across cortical hierarchies in real time. *Nature Machine Intelligence*, 6(6), 619–630.
- [33] Ma, C., Wu, J., Si, C., & Tan, K. C. (2024). Scaling supervised local learning with augmented auxiliary networks. *International Conference on Learning Representations*.
- [34] Lv, C., Xu, J., Lu, Y., Wang, X., Wang, Z., Xu, Z., Yu, D., Du, X., Zheng, X., & Huang, X. (2025). Dendritic Localized Learning: Toward Biologically Plausible Algorithm. *Proceedings of the 42nd International Conference on Machine Learning, Proceedings of Machine Learning Research*, 267, 41682–41700.
- [35] Erdogan, M., Pehlevan, C., & Erdogan, A. T. (2025). Error broadcast and decorrelation as a potential artificial and natural learning mechanism. *NeurIPS 2025 Spotlight*, OpenReview.
- [36] Kao, C.-H., & Hariharan, B. (2024). Counter-Current Learning: A biologically plausible dual network approach for deep learning. *Advances in Neural Information Processing Systems*, 37.
- [37] Xiao, H., Rasul, K., & Vollgraf, R. (2017). Fashion-MNIST: A novel image dataset for benchmarking machine learning algorithms. *arXiv:1708.07747*.
- [38] Song, S., Sjöström, P. J., Reigl, M., Nelson, S., & Chklovskii, D. B. (2005). Highly Nonrandom Features of Synaptic Connectivity in Local Cortical Circuits. *PLoS Biology*, 3(3), e68.
- [39] Buzsáki, G., & Mizuseki, K. (2014). The log-dynamic brain: How skewed distributions affect network operations. *Nature Reviews Neuroscience*, 15(4), 264–278.
- [40] The MICrONS Consortium et al. (2025). Functional connectomics spanning multiple areas of mouse visual cortex. *Nature*, 640(8058), 435–447.

Dataset	Rule	Top-10 valid	Top-10 test
MNIST	3F	0.611	0.622
MNIST	4F	0.620	0.628
MNIST	5F	0.912	0.916
Context gating	3F	0.398	0.396
Context gating	4F	0.411	0.411
Context gating	5F	0.807	0.789

Table S1: **Rule-family ranking.** Top-10 mean across completed local-competence sweeps.

- [41] Loewenstein, Y., Kuras, A., & Rumpel, S. (2011). Multiplicative Dynamics Underlie the Emergence of the Log-Normal Distribution of Spine Sizes in the Neocortex In Vivo. *Journal of Neuroscience*, 31(26), 9481–9488.
- [42] Megías, M., et al. (2001). Total number and distribution of inhibitory and excitatory synapses on hippocampal CA1 pyramidal cells. *Neuroscience*, 102(3), 527–540.
- [43] Francioni, V., Tang, V. D., Toloza, E. H. S., Ding, Z., Brown, N. J., & Harnett, M. T. (2026). Vectorized instructive signals in cortical dendrites. *Nature*. <https://doi.org/10.1038/s41586-026-10190-7>
- [44] Iyer, A., Grewal, K., Velu, A., Souza, L. O., Forest, J., & Ahmad, S. (2022). Avoiding catastrophe: Active dendrites enable multi-task learning in dynamic environments. *Frontiers in Neurorobotics*, 16, 846219.

A Supplementary Results

Rule-Family Ranking

Which Components Carry?

Broadcast Mode Comparison and Local-Mismatch Recheck

Phase 1 Capacity Ceilings

Local Competence and Regime Dependence

Ablation Results

Extended Gradient Analysis and N_I Sweep Detail

Verification and Reproducibility

Additional Stress Tests

FA/DFA Baseline Comparison

CIFAR-10 Results

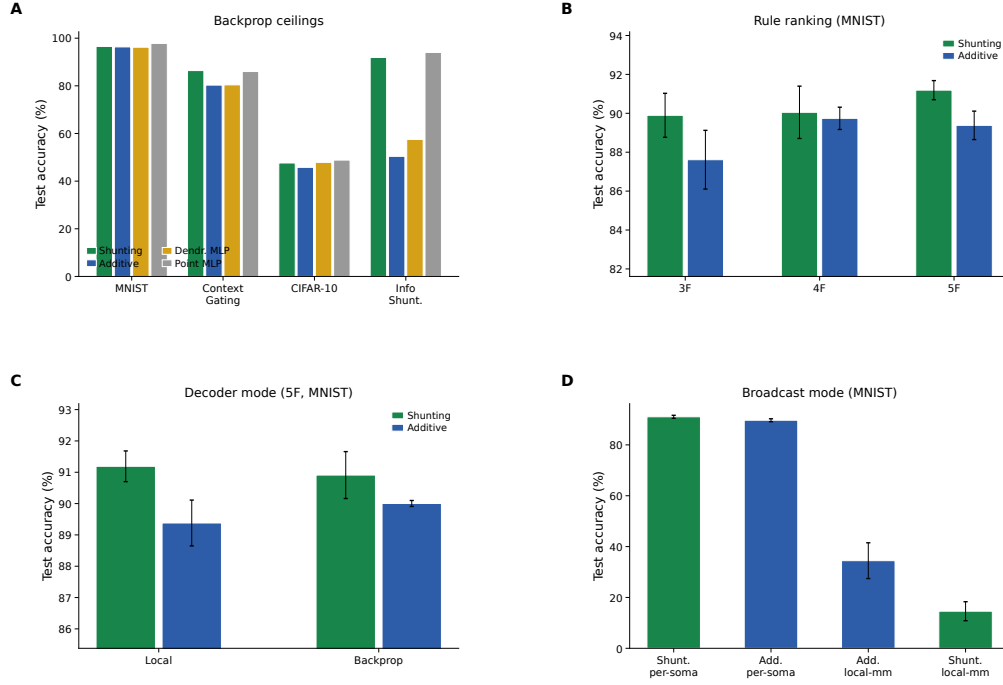


Figure S1: **Capacity calibration (supplementary).** (A) Phase 1 backprop ceilings across all architectures and datasets. (B) Rule-family ranking: 5F consistently outperforms 4F and 3F. (C) Decoder mode comparison (local vs. backprop decoder, 5F MNIST). (D) Broadcast mode comparison: per-soma is required for strong performance; local-mismatch fails.

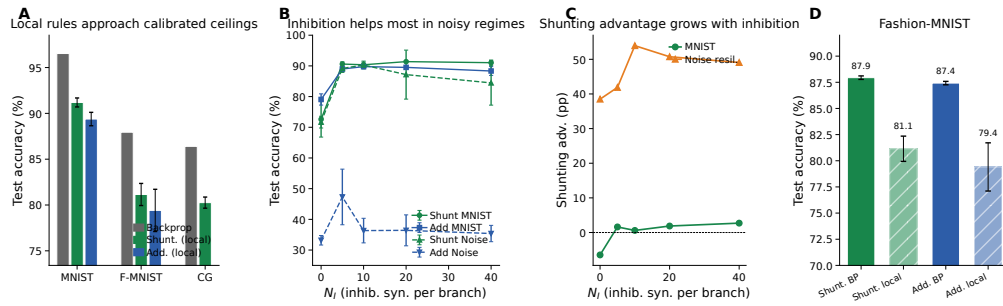


Figure S2: **Local competence and regime dependence.** (A) Backprop ceiling (gray) vs. best local rule (5F per-soma) for shunting (green) and additive (blue) cores on MNIST, Fashion-MNIST, and context gating. (B) N_I dose-response: test accuracy vs. inhibitory synapses per branch (N_I) for shunting and additive cores on MNIST (solid) and noise-resilience (dashed). Shunting needs $N_I \geq 5$ to reach high accuracy; additive learning fails on noise-resilience regardless of N_I . (C) Shunting advantage (Δ , in percentage points) vs. N_I . The shunting advantage grows with inhibitory synapses per branch and is larger on the noise-resilience task than on MNIST. (D) Fashion-MNIST.

Component	Representative effect	Interpretation
Shunting architecture	Conductance-stage path-gain CV stays at about 0.21–0.27 under shunting vs. 0.53–1.08 under additive; on noise resilience, per-soma local learning rises from roughly 0.33–0.47 (additive) to 0.72–0.90 (shunting).	This is the main architectural effect in the paper: shunting concentrates conductance-stage path gains and yields a more faithful low-bandwidth credit signal.
Broadcast quality	On MNIST (5F), shunting per-soma reaches 0.912 ± 0.005 , while local-mismatch remains at 0.146 ± 0.046 .	Correctly matching the broadcast field to the somatic error matters far more than arbitrary local mismatch heuristics.
Transported oracle	On noise resilience, additive local learning rises from 33–47% under per-soma to 75.6–98.5% under transported broadcast; shunting rises from 71.7–90.3% to 98.5–99.3%.	The oracle is the strongest causal evidence that broadcast fidelity, not just architecture, is the dominant bottleneck.
Rule hierarchy	In the competence sweeps, MNIST improves from 0.622 (3F) and 0.628 (4F) to 0.916 (5F); context gating improves from 0.396/0.411 to 0.789.	5F is the strongest practical rule in this paper, though its extra factor remains heuristic rather than theorem-derived.
5F stability	Tightening the ϕ clamp to $[0.5, 2.0]$ changes MNIST by only about +0.4 pp; varying EMA from 0.05 to 0.20 changes it by less than 0.2 pp.	The practical gain from 5F is not especially brittle to its main tuning knobs.
Structured feedback	On learned cue routing, tuned shunting per-soma reaches $88.2 \pm 13.9\%$, while tuned pathway-vector feedback reaches $99.1 \pm 0.3\%$.	Structured feedback matters when different branches must be credited differently; it is not a generic replacement for per-soma broadcast on ordinary tasks.
Random low-rank channels	On noise resilience, random low-rank broadcast improves from 0.456 at $K=1$ to 0.606 at $K=4$, but remains far below the stronger per-soma and transported-broadcast baselines; on cue routing, raising random rank from $K=1$ to $K=2$ does not match PV-LocalCA.	Dimensionality alone is not enough: pathway structure matters, not just additional random broadcast channels.
Morphology regime	In the exploratory noise-resilience regime map (Appendix Fig. S10), depth-2 trees show their largest shunting gains near $N_I = 0$, whereas depth-3 trees retain their strongest gains around $N_I = 5$.	Morphology appears to shift the inhibitory operating regime in which shunting is most useful, suggesting that conductance normalization and tree geometry interact rather than contributing independently.
Activation choice	In the five-seed frozen activation audit, identity favors additive LocalCA on noise resilience (0.662/0.647 at $N_I=0/10$), whereas frozen <code>param_tanh</code> favors shunting (0.657/0.679). The same audit shows that frozen <code>param_tanh</code> improves shunting per-soma compartment-error cosine from 0.124 to 0.329 at $N_I=0$.	The post-voltage firing-rate transform is a real modeling variable, not just an implementation detail. Even a fixed saturating nonlinearity changes the operating point substantially, so the main paper’s mechanistic claims should be read in the bounded <code>param_tanh</code> regime used throughout the headline experiments.

Table S2: **Which components carry the observed gains?** A compact summary of the empirical division of labor across architecture, broadcast design, and LocalCA rule variants.

Core	Broadcast	Decoder	Test (mean $\pm$ std)
Shunting	per-soma	local	0.912 $\pm$ 0.005
Shunting	per-soma	backprop	0.909 $\pm$ 0.008
Shunting	local-mismatch	local	0.146 $\pm$ 0.046
Shunting	local-mismatch	backprop	0.146 $\pm$ 0.037
Additive	per-soma	local	0.894 $\pm$ 0.007
Additive	per-soma	backprop	0.900 $\pm$ 0.001
Additive	local-mismatch	local	0.342 $\pm$ 0.058
Additive	local-mismatch	backprop	0.348 $\pm$ 0.095

Table S3: **Local-mismatch recheck (MNIST, 5F)**. Per-soma is consistently strong; local-mismatch remains much weaker.

Dataset	BP ceiling	Best local (5F)	Gap
MNIST	0.965	0.914 $\pm$ 0.003	5.1%
Fashion-MNIST	0.879	0.811 $\pm$ 0.012	6.8%
Context gating	0.864	0.803 $\pm$ 0.006	6.1%

Table S4: **Local competence**. Backprop ceilings are matched-capacity selection estimates: MNIST and context gating use the best phase-1 standard-training setting from a 2-seed capacity-calibration sweep, whereas Fashion-MNIST uses a dedicated 5-seed competence sweep. Local values are 5F with per-soma broadcast on shunting dendritic cores. Context gating additionally uses an HSIC auxiliary objective (weight 0.01; Appendix E). Error bars on local values: ± 1 s.d. across 5 seeds.

Condition	Metric	Value
Decoder: local vs backprop vs frozen (MNIST, sandbox)	test acc	0.379 vs 0.379 vs 0.176
Shunting vs additive, matched (MNIST, sandbox)	Δ test	+0.210
Per-soma, path on vs off (synthetic task, sandbox)	test / MI(E,I;C)	-0.003 / +0.017

Table S5: **Mechanistic ablations**. Results from controlled small-network sandbox experiments using minimal architectures to isolate individual factors. Sandbox runs use a small 2-layer dendritic network (branch factor [9]) trained for 30 epochs; they are not directly comparable to the headline capacity-calibrated results.

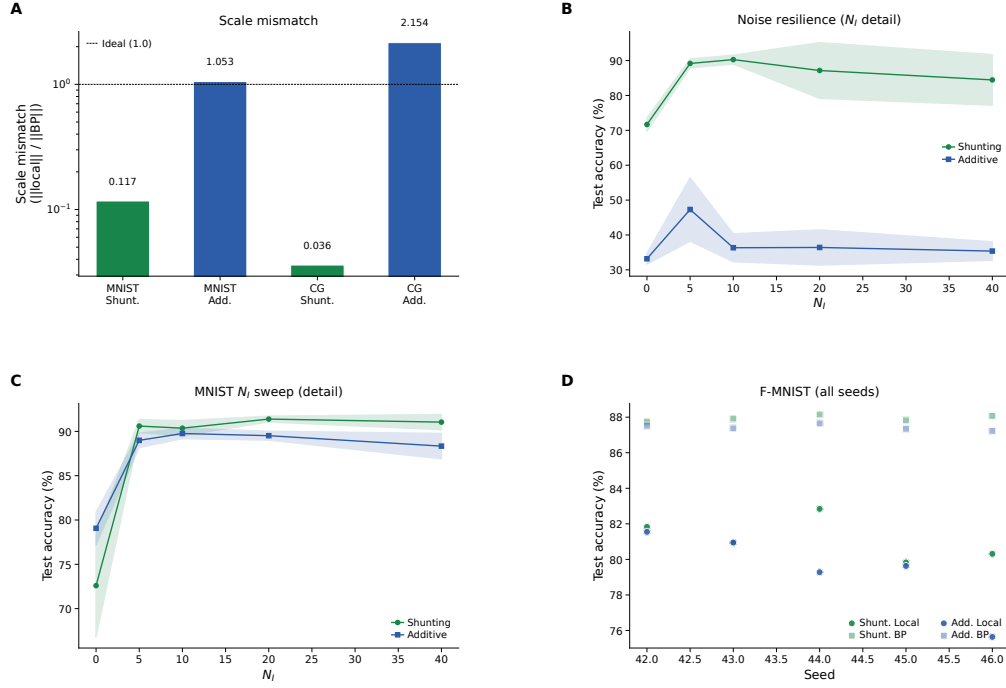


Figure S3: **Extended gradient and N_I analysis (supplementary).** (A) Scale mismatch bars: shunting achieves near-ideal scale (0.117); additive exhibits order-of-magnitude distortion (> 1.0). (B) Noise-resilience N_I dose-response with error bands (± 1 s.d.). (C) MNIST N_I dose-response detail with error bands. (D) Fashion-MNIST individual seeds for all conditions, showing consistency across runs.

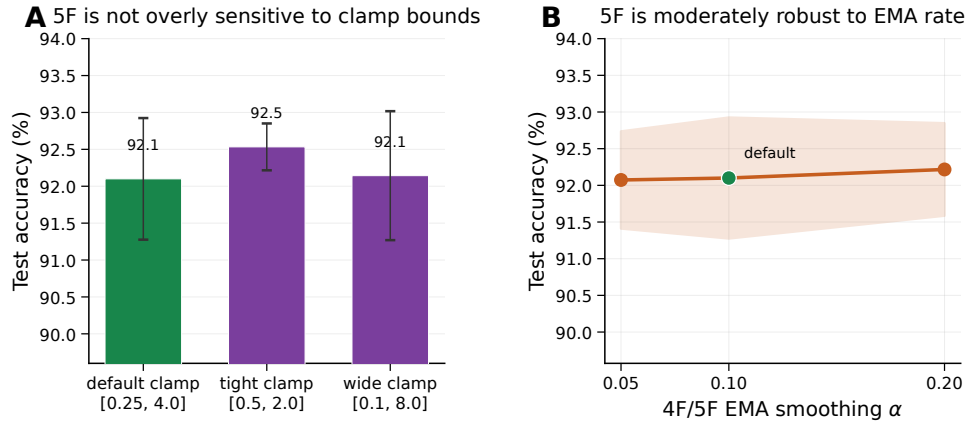


Figure S4: **5F sensitivity analysis (MNIST, shunting, per-soma broadcast).** (A) Sensitivity to the ϕ_n clamp interval. The default clamp [0.25, 4.0] is already strong ($92.1 \pm 0.8\%$); tightening it to [0.5, 2.0] slightly improves performance ($92.5 \pm 0.3\%$), while widening it to [0.1, 8.0] is effectively neutral ($92.1 \pm 0.9\%$). (B) Sensitivity to the 4F/5F EMA rate used for online statistics. Varying α from 0.05 to 0.20 changes mean test accuracy by less than 0.2 pp. Together these results suggest that the 5F gain is not driven by a brittle single hyperparameter choice, even though the factor itself remains heuristic. Errors: ± 1 s.d. across 3 seeds.

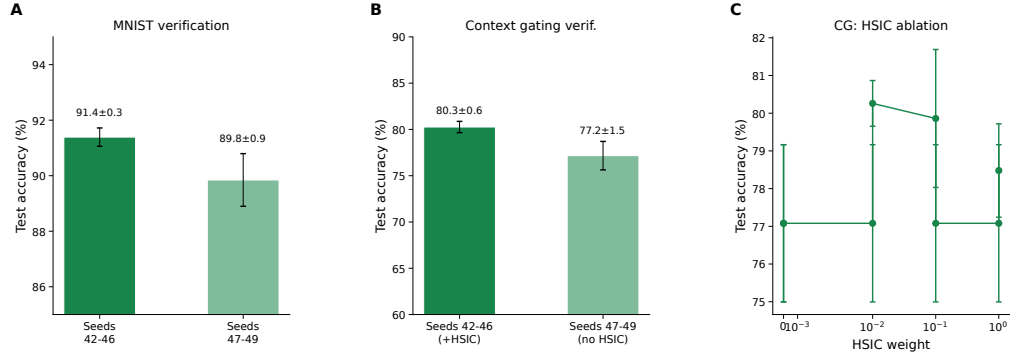


Figure S5: **Verification and reproducibility (supplementary).** (A) MNIST verification: main seeds (42–46) yield $91.4 \pm 0.3\%$; held-out seeds (47–49) yield $89.8 \pm 0.9\%$, confirming generalization. (B) Context gating verification: main seeds (+HSIC) $80.3 \pm 0.6\%$; held-out seeds (no HSIC) $77.2 \pm 1.5\%$. The roughly 3 pp gap reflects HSIC removal, not seed sensitivity. (C) HSIC weight ablation on context gating: moderate weights (0.01–0.1) perform best.

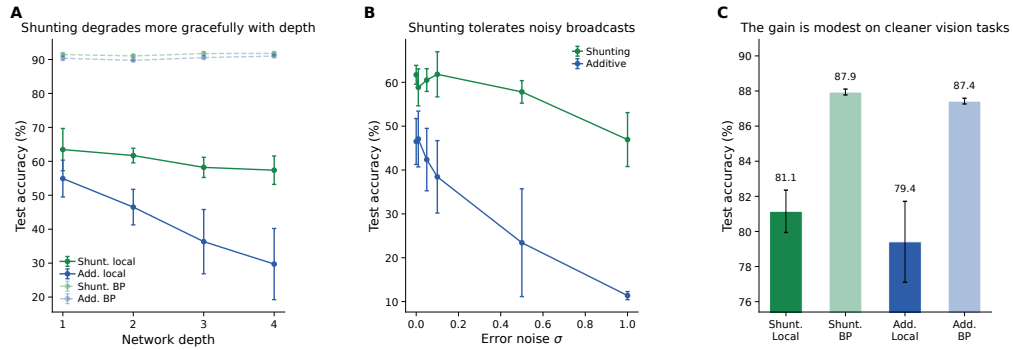


Figure S6: **Additional stress tests.** (A) Depth scaling: shunting local (green) degrades more gracefully from 63.5% to 57.4%; additive local (blue) drops from 54.9% to 29.7%. Backprop ceilings (dashed) remain stable. (B) Broadcast-noise robustness: shunting remains stable for $\sigma \leq 0.1$; additive degrades rapidly and reaches chance at $\sigma = 1$. (C) Fashion-MNIST: the shunting gain is modest on this cleaner benchmark (81.1% local vs. 87.9% backprop), consistent with the regime-dependent story in the main text.

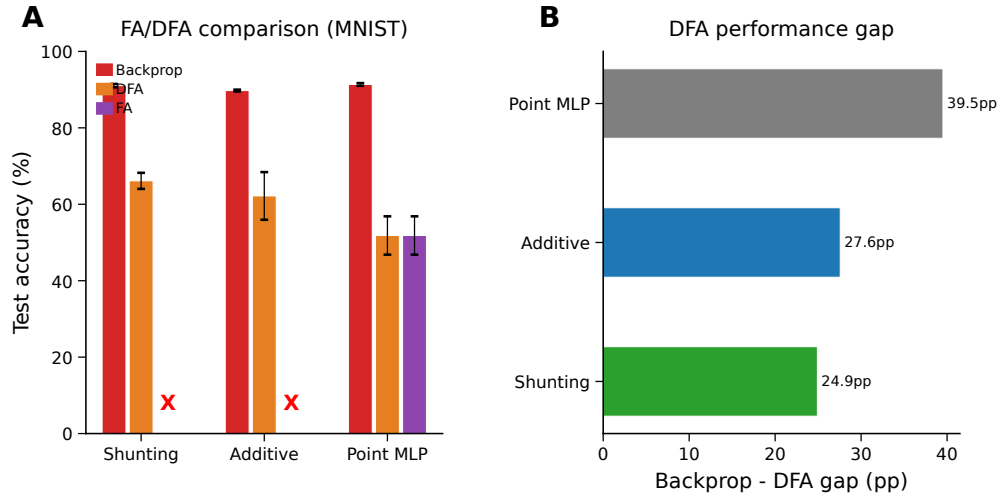


Figure S7: **Feedback alignment baselines (MNIST)**. (A) Grouped comparison: standard backprop (gray), DFA (orange), and FA (purple). Red X marks indicate FA failure on dendritic architectures (dimensional incompatibility between random feedback matrices and block-structured dendritic layers). DFA achieves 66.1% on shunting vs. 62.2% additive vs. 51.8% point MLP. (B) Backprop–DFA gap: dendritic architectures (24.9–27.6 pp) show smaller gaps than point MLPs (39.5 pp), suggesting conductance-based architecture is partially compatible with random feedback.

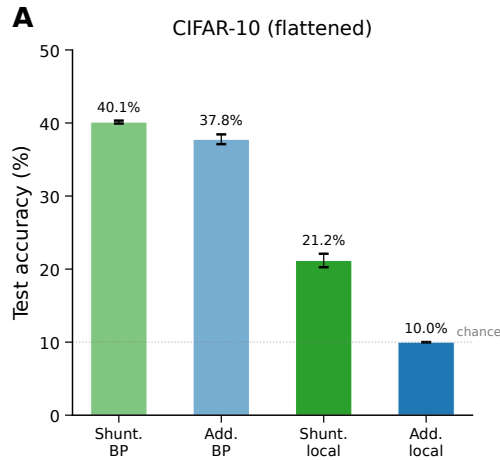


Figure S8: **Conservative flattened CIFAR-10 reference sweep**. In this appendix stress test, shunting backprop reaches 40.1%, additive backprop 37.8%, shunting local 21.2%, and additive local 10.0% (chance). The backprop–local gap (about 19 pp for shunting) is larger than on MNIST, reflecting the increased difficulty of propagating credit through conductance-based layers for complex visual features. This panel should be read as a conservative flattened-image benchmark rather than as the tuned CIFAR ceiling for the architecture family.

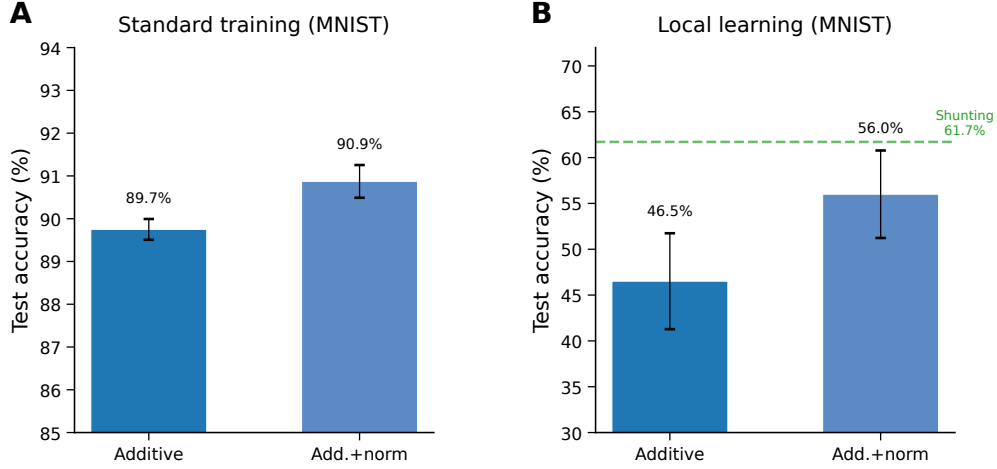


Figure S9: **Additive + normalization control (MNIST)**. (A) Standard backprop: normalization provides a small boost (89.7% $\rightarrow$ 90.9%). (B) Local learning: normalization improves additive from 46.5% to 56.0% (+9.5 pp), partially closing the gap to shunting (61.7%, dashed green). The remaining gap (−5.7 pp) indicates that shunting provides benefits beyond simple voltage normalization.

Broadcast rank K	Noise resilience	Cue routing
1	0.456 ± 0.111	0.877 ± 0.084
2	0.547 ± 0.097	0.839 ± 0.106
4	0.606 ± 0.044	—
8	0.546 ± 0.100	—
10	0.563 ± 0.104	—

Table S6: **Random low-rank broadcast does not replace structure**. On noise-resilience, a small number of random broadcast channels helps somewhat, with a weak optimum near $K=4$, but remains far below the stronger per-soma and transported-broadcast baselines in the main text. On cue routing, increasing unstructured rank from $K=1$ to $K=2$ does not recover the pathway-aware gains of PV-LocalCA.

438 Additive Normalization Control

439 Morphology $\times$ Inhibition Regime Map

440 Random Low-Rank Broadcast Channels

441 Frozen Activation-Choice Audit

442 B Implementation Details

443 Biological Plausibility Assumptions

Each synapse has access to: presynaptic activity x_j (local), compartment voltage V_n (local membrane potential), reversal potential E_j (fixed by receptor type), and total input resistance $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$ (computable from local conductances). The 4F modulator ρ_n requires online estimation of voltage covariance (biologically plausible via slow calcium signals); the 5F factor ϕ_n requires a linear regression proxy (implementable via eligibility traces). The only non-local quantity is the broadcast error e_n , which requires a top-down or neuromodulatory signal from the soma to dendritic compartments. We assume per-neuron resolution (one error per output neuron) for the per-soma broadcast mode.

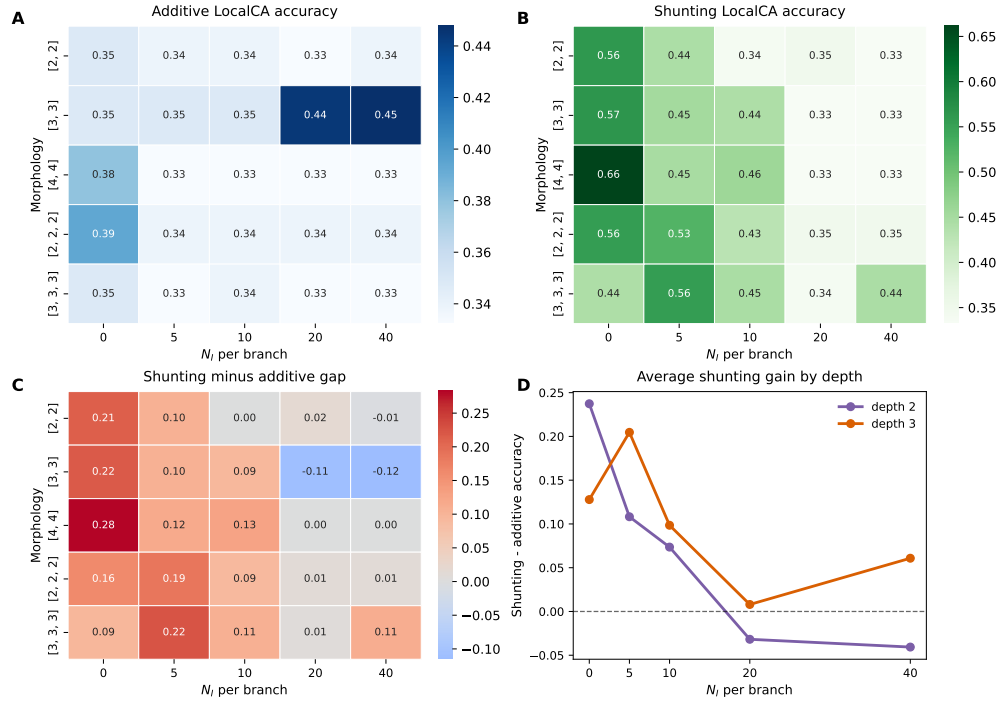


Figure S10: **Exploratory morphology × inhibition regime map on noise resilience.** (A) Additive LocalCA remains comparatively flat across morphologies and inhibitory synapse counts, with only isolated high-variance pockets. (B) Shunting LocalCA shows a clear morphology-dependent regime: the strongest performance appears at low or moderate inhibition, and the best N_I shifts with tree shape. (C) The shunting-minus-additive gap is largest for wider two-level trees at low inhibition and for deeper trees at modest inhibition, then weakens or vanishes at high inhibition. (D) Averaging by depth highlights the same shift: depth-2 trees peak near $N_I = 0$, whereas depth-3 trees retain their largest shunting advantage around $N_I = 5$. We treat this figure as exploratory support for the claim that morphology changes the inhibitory operating regime in which shunting is most useful; a fuller mechanistic account would require matched theory diagnostics for each cell in the grid.

451 Dendritic Structure in Code

452 We use the morphology notation $[b_1, b_2, \dots, b_D]$ for rooted dendritic trees, where each factor gives
 453 the fan-in from one dendritic stage to the next on the path to the soma. For example, $[3, 3]$ means that
 454 each soma aggregates 3 proximal branches and each proximal branch aggregates 3 distal branches,
 455 for 9 distal leaves per soma; $[3, 3, 3]$ adds a third dendritic stage with 27 leaves per soma. In code,
 456 each excitatory unit is instantiated as a DendriNet, which is a stack of DendriticBranchLayers.
 457 Each branch layer contains an excitatory synapse bank, an optional inhibitory synapse bank, a
 458 branch-to-parent aggregation map, and an optional post-voltage reactivation. In the main feed-
 459 forward models studied here, `somatic_synapses=false`, so synaptic inputs terminate on den-
 460 dritic branches rather than directly on the soma. Likewise, `input_mode=1` together with nonzero
 461 `ie_synapses_per_branch_per_layer` means that inhibition enters each dendritic branch through
 462 explicit inhibitory synapses even when we do not instantiate a separate inhibitory neuron popula-
 463 tion (`inhibitory_layer_sizes=[]`). The additive and shunting cores are therefore architecture-
 464 matched at the tree level; they differ in the branch-level voltage integration rule rather than in whether
 465 a dendritic tree exists.

466 Post-Voltage Reactivation

467 The conductance stage produces a pre-activation compartment voltage V_n , after which each branch
 468 layer may apply a monotone scalar nonlinearity. The codebase supports identity (none), fixed
 469 `tanh/sigmoid/relu`, and several parametric families including `param_tanh`, `param_tanh_only_m`,

Dataset / strategy	Core	N_I	Identity (none)	Frozen param_tanh
MNIST / LocalCA	additive	0	0.780 ± 0.014	0.184 ± 0.049
MNIST / LocalCA	additive	10	0.907 ± 0.003	0.841 ± 0.011
MNIST / LocalCA	shunting	0	0.841 ± 0.007	0.774 ± 0.023
MNIST / LocalCA	shunting	10	0.905 ± 0.009	0.846 ± 0.015
Noise resilience / LocalCA	additive	0	0.662 ± 0.015	0.608 ± 0.022
Noise resilience / LocalCA	additive	10	0.647 ± 0.010	0.466 ± 0.034
Noise resilience / LocalCA	shunting	0	0.369 ± 0.011	0.657 ± 0.018
Noise resilience / LocalCA	shunting	10	0.368 ± 0.015	0.679 ± 0.020
MNIST / backprop	additive	0	0.925 ± 0.001	0.942 ± 0.001
MNIST / backprop	additive	10	0.921 ± 0.001	0.950 ± 0.001
MNIST / backprop	shunting	0	0.963 ± 0.0004	0.950 ± 0.001
MNIST / backprop	shunting	10	0.967 ± 0.001	0.946 ± 0.001
Noise resilience / backprop	additive	0	0.649 ± 0.015	0.995 ± 0.002
Noise resilience / backprop	additive	10	0.646 ± 0.003	0.991 ± 0.005
Noise resilience / backprop	shunting	0	0.361 ± 0.015	0.494 ± 0.015
Noise resilience / backprop	shunting	10	0.386 ± 0.024	0.885 ± 0.006

Table S7: **Frozen activation-shape audit on matched dendritic architectures (5 seeds per condition)**. Identity (none) and a frozen bounded post-voltage reactivation param_tanh can lead to qualitatively different operating regimes. On the main synthetic mechanism task, identity favors additive LocalCA, whereas frozen param_tanh strongly favors shunting. Under standard backpropagation, the same frozen nonlinearity markedly improves additive and helps shunting most clearly at higher inhibition. These runs match additive reactivation initialization to the shunting setting and hold the activation parameters fixed, isolating activation shape from both the earlier additive/shunting initialization asymmetry in code and from activation learning itself.

Dataset	Core	N_I	Path-gain CV none	Path-gain CV Frozen param_tanh	Per-soma cosine none	Per-soma cosine Frozen param_tanh	Path-transport cosine none	Path-transport cosine Frozen param_tanh
MNIST	additive	0	0.281	0.330	0.229	0.006	0.996	0.670
MNIST	additive	10	0.310	0.863	0.210	0.074	0.963	0.222
MNIST	shunting	0	0.383	0.793	0.167	0.111	1.000	0.964
MNIST	shunting	10	0.386	1.246	0.235	0.097	1.000	0.966
Noise resilience	additive	0	0.146	0.229	0.478	-0.250	0.987	0.780
Noise resilience	additive	10	0.239	0.285	0.780	0.009	0.974	0.347
Noise resilience	shunting	0	0.299	0.316	0.124	0.329	0.315	0.678
Noise resilience	shunting	10	0.041	0.160	0.352	0.386	0.999	0.889

Table S8: **Pre-reactivation theory diagnostics for the frozen activation audit (LocalCA, 5 seeds per condition)**. These diagnostics are computed on the hooked pre-reactivation compartment voltage V_n , so they track how the fixed post-voltage activation changes the operating point seen by the conductance-stage theory. The strongest qualitative pattern appears on the main synthetic mechanism task: frozen param_tanh improves shunting per-soma compartment-error fidelity but reduces it sharply in additive cores. We present these results as an operating-point audit rather than as a second theorem claim; the exact factorization statement in the main text remains the conductance-stage result verified in Fig. 2D.

and piecewise linear-saturating variants. The main manuscript sweeps use param_tanh, a learnable bounded firing-rate transform of the form $(\tanh(m(V - b)) + 1)/2$, with identity transfer input activation and a linear decoder. Unless explicitly overridden, additive and shunting cores can start from different reactivation initializations in code; activation-fairness ablations therefore match additive initialization to the shunting setting when comparing alternative reactivation choices directly. The appendix activation audit additionally fixes param_tanh after initialization to isolate activation shape from activation plasticity.

Units and Parameterization

Decoder Update Modes

W_{dec} maps $V_L \rightarrow \hat{y}$. Modes: **backprop** ($\nabla_W L$ via autograd), **local** ($\Delta W = \eta \langle \delta_0 V_L^\top \rangle_B$), **frozen** ($\Delta W = 0$).

Quantity	Symbol	Convention
Voltage	V	Normalized to $[-1, 1]$
Conductances	$g^{\text{syn}}, g^{\text{den}}$	Nonneg. via softplus
Leak conductance	g^{leak}	Set to 1
Input resistance	R^{tot}	≤ 1

Table S9: Units and normalization.

Category	Implemented modes	Role in this paper
Core models	point MLP; dendritic MLP; dendritic additive; dendritic shunting; pathway-routed dendritic cores	Main comparisons use dendritic additive vs. dendritic shunting; routed cores are used for cue integration.
Training strategies	standard backprop; LocalCA; DFA; FA; transported-broadcast oracle	Main text uses standard and LocalCA; DFA/FA remain appendix baselines; path transport is an oracle upper bound.
Broadcast modes	scalar; per-soma; local mismatch; low-rank; path transport; pathway vector	Main text centers on per-soma and pathway-vector; scalar and local-mismatch are supporting controls; path transport is diagnostic only.
Recent dendritic extensions	path propagation; router-derived branch roles; pathway-vector feedback; inhibitory homeostasis	These were added to test morphology-aware credit transport and structured branch-specific plasticity beyond the original 3F/4F/5F family.

Table S10: Implemented model families and LocalCA operating modes. Only a subset is used for the main claims; the remaining modes serve as controls, ablations, or structured extensions.

Model and Mode Inventory

Representative Manuscript Architectures

LocalCA Extension Compatibility

Seed Protocol and Claim Status

The seed depth is not uniform across every sweep, and we therefore separate headline evidence from exploratory screening. The main mechanistic gradient-fidelity sweep uses 5 seeds per condition, the cue-routing seed-repeat and tuning sweeps use 5 seeds per condition, the dedicated Fashion-MNIST competence sweep uses 5 seeds per condition, the path-transport oracle sweep uses 3 seeds per condition, the 5F sensitivity sweep uses 3 seeds per condition, and the frozen activation audit in Tables S7–S8 uses 5 seeds per condition. By contrast, early architecture-selection or stress-test screens with only 1–2 seeds are treated as exploratory and are either omitted from the main claims or explicitly framed as appendix-only pilots. For transparency, the BP ceilings in Table S4 for MNIST and context gating come from a matched-capacity phase-1 calibration sweep with 2 seeds, whereas the corresponding local runs use 5 seeds and the Fashion-MNIST standard/local comparison uses a dedicated 5-seed sweep. This asymmetry affects the confidence we should place in the ceiling estimates, not the mechanistic sweeps that anchor the paper’s main claims.

Compute Resources

All experiments were run on an internal GPU cluster. Most MNIST, Fashion-MNIST, and synthetic-task runs used a single NVIDIA A100 or H100 GPU and finished in minutes to tens of minutes per seed; the flattened CIFAR-10 runs required longer single-GPU jobs but still did not use distributed or multi-GPU training. Reported headline metrics are aggregated over 3–5 seeds as stated in the text; the broader exploratory sweeps used to tune and diagnose the method required additional compute beyond the final reported runs.

Setting	Encoder	E layers	Tree	Synapses / branch	Seeds	Notes
Gradient fidelity / path-gain / noise-resilience mechanism	identity	[128]	[3, 3]	$N_E=40, N_I \in \{0, 5, 10, 20, 40\}$	5 fidelity, 3 oracle	Main mechanistic sweeps in Figs. 2 and 3; dendritic reactivation is <code>param_tanh</code> .
MNIST / context-gating local competence	identity	[128]	[3, 3]	$N_E=40, N_I=20$	5	Best local 5F per-soma runs used in Table S4; dendritic reactivation is <code>param_tanh</code> .
Fashion-MNIST competence	identity	[128]	[3, 3]	$N_E=40, N_I=20$	5	Dedicated five-seed standard-vs-local competence sweep; dendritic reactivation is <code>param_tanh</code> .
Cue-routing PV-LocalCA	pathway router	[64]	[2]	$N_E=10, N_I=4$	5	Learned router with two pathway groups and rank-2 structured feedback; dendritic reactivation is <code>param_tanh</code> .
Morphology×inhibition appendix map	identity	[128, 128]	[2, 2], [3, 3], [4, 4], [2, 2, 2], [3, 3, 3]	$N_E=40, N_I \in \{0, 5, 10, 20, 40\}$	3	Exploratory appendix sweep only; dendritic reactivation is <code>param_tanh</code> .

Table S11: Representative architectures used in the manuscript. All dendritic rows use explicit branch-level excitatory and inhibitory synapse banks with `somatic_synapses=false` and nonnegative conductances. The tree column gives the dendritic morphology per excitatory neuron, while the synapse counts report excitatory and inhibitory synapses per branch.

Extension	Meaning	Compatible with	Typical use here
Per-soma broadcast	Somatic error vector shared across compartments	3F, 4F, 5F	Default main-text LocalCA setting
Low-rank broadcast	Small number of broadcast channels instead of a scalar or full field	3F, 4F, 5F	Intermediate control when rank-1 is too restrictive
Path transport	Recursive tree-transported error using conductance-stage path gains	3F, 4F, 5F	Oracle upper-bound broadcast
Path propagation	Approximate path-gain attenuation applied to broadcast	3F, 4F, 5F	Morphology-aware modifier for deeper trees
Pathway vector	Structured branch/pathway-specific broadcast field	3F, 4F, 5F	Used mainly in PV-LocalCA on routed tasks
Branch-role rules	Scales updates by router- or pathway-derived branch specialization	3F, 4F, 5F	Used in the structured pathway extension
Inhibitory homeostasis	Auxiliary local inhibitory balancing term	3F, 4F, 5F	Optional stabilizer in routed or shunting settings

Table S12: Compatibility of the newer LocalCA extensions with the base 3F/4F/5F family. These extensions mostly modify the error field or branch scaling rather than defining a new “6F” or “7F” rule.

504 Code, Data, and Asset Availability

505 MNIST, Fashion-MNIST, CIFAR-10, and MICrONS are public datasets or public benchmark assets
506 cited in the paper; we use them under their standard published access conditions and cite their original
507 sources in the bibliography. The present anonymous submission does not include a public code URL
508 in order to preserve double-blind review, but the paper specifies the model family, training modes, key
509 hyperparameters, and representative configurations, and we plan to release code and configuration
510 files after review. The paper does not release a new dataset or a stand-alone pretrained model asset.

Rule	Factors	Cost	Best regime
3F	$x, (E - V), e$	$\mathcal{O}(1)$	Baseline
4F	$3F + \rho$	$\mathcal{O}(1)$	Improved conditioning
5F	$4F + \phi$	$\mathcal{O}(d_n)$	Strongest overall
5F+VH	5F + local voltage homeostasis	$\mathcal{O}(d_n)$	Stabilized LocalCA / warm-up
PV-LocalCA	3F + structured e_n + role alignment	$\mathcal{O}(r_n)$	Pathway discovery

Table S13: Variant taxonomy. ‘5F+VH’ denotes the same base 5F rule with an additional strictly local voltage-centering auxiliary that can also be scheduled through multi-stage LocalCA warm-up; the codebase also supports this as a shorthand ‘5f_vh’ alias in configuration files.

Component	Analog	Interpretation
R_n^{tot}	Input resistance	Sensitivity modulation
$(E_j - V_n)$	Synaptic driving force	Local gradient factor
Shunting	Divisive normalization	$\partial V / \partial g_I \propto -V$
ρ_n	Layer relevance	Output correlation
ϕ_n	Signal propagation	Conditional predictability
Pathway vector	Branch-specific top-down drive	Structured teaching signal
Role alignment	Pathway-selective plasticity	Branch specialization

Table S14: Biological analogs.

511 Algorithm

Algorithm 1 Local Credit Assignment

- 1: **Input:** Model, batch (x, y) , config $\mathcal{C}$
- 2: Forward pass; loss L , output error δ^y
- 3: Somatic error $\delta_0 = W_{\text{dec}}^\top \delta^y$
- 4: **for** each layer n (reverse) **do**
- 5: $e_n = \text{broadcast}(\delta_0, \mathcal{C})$; optionally gate by pathway roles
- 6: Compute ρ_n, ϕ_n (EMA estimators, if enabled)
- 7: Apply the configured local update (Eq. 6 or 7)
- 8: **end for**
- 9: Clip gradients; optimizer step

512 C Theoretical Details

513 Variant Taxonomy

514 Random-Broadcast Alignment

515 **Proposition 3** (Supportive random-broadcast alignment). *Let the broadcast matrix B_n have i.i.d.*
516 *zero-mean entries with $\mathbb{E}[B_n^\top B_n] = \alpha I$. Then the expected cosine between the resulting local*
517 *gradient and the exact gradient is positive, $\mathbb{E}[\cos \angle(g^{\text{local}}, g^{\text{exact}})] \geq c_n > 0$, by an argument*
518 *analogous to feedback alignment [9]. The constant c_n depends on how strongly local eligibility*
519 *factors correlate with the conductance-stage path gain in Eq. (4); shunting improves this correlation*
520 *by narrowing the spread of intermediate sensitivities.*

521 We treat this argument as supportive rather than central. The main theoretical object in the paper
522 is still the exact pre-reactivation compartment error $\partial L / \partial V_n$; under identity upward transfer this
523 reduces to $\alpha_n \delta_0$, while with post-voltage activation the corresponding effective path gain additionally
524 includes local $f'_n(V_n)$ factors. The main empirical question is how faithfully different broadcast fields
525 approximate that quantity.

526 Biological Analogs

527 D Morphology-Aware Extensions

528 **Path-integrated propagation.** Modulate broadcast error by $\pi_n = \pi_{n-1} \cdot R_{n-1}^{\text{tot}} \cdot \bar{g}_{n-1}^{\text{den}}$, approxi-
529 mating depth attenuation from Eq. 4.

530 **Depth modulation.** Per-branch scaling $\rho_j = \rho_{\text{base}} / (d_j + \alpha)$, mirroring cable attenuation.

531 **Dendritic normalization.** $\Delta g_j^{\text{den}} \leftarrow \Delta g_j^{\text{den}} / (\sum_k g_k^{\text{den}} + \varepsilon)$, analogous to homeostatic scaling
532 [21].

533 **Pathway-vector broadcast.** For tasks with latent pathway structure, the broadcast becomes a
534 low-dimensional vector over router-inferred pathway roles. This vector is gated by local pathway
535 activity and propagated through the block-structured tree so that earlier branches receive feedback
536 aligned with their downstream descendants.

537 **Router-derived branch roles.** Each branch receives a role profile inferred from router assignments
538 and incoming block weights rather than from a hand-coded apical/basal label. Plasticity is then
539 modulated by branch selectivity and a synapse-specific role-alignment factor, emphasizing pathway-
540 consistent updates without imposing a heuristic branch taxonomy.

541 E HSIC Auxiliary Objectives

542 Following [17], we use kernel-based HSIC objectives. Self-decorrelation: $\mathcal{L}^{\text{self}} =$
543 $B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Z \mathbf{H})$. Target-correlation: $\mathcal{L}^{\text{target}} = -B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Y \mathbf{H})$. Moderate weights
544 (0.01–0.1) help on context gating; negligible on MNIST. Online statistics (ρ_n, ϕ_n) use Welford’s
545 algorithm [20].

546 F Online Variant with Eligibility Traces

547 Continuous-time eligibility: $\tau_e \dot{e}_j^{\text{syn}} = -e_j^{\text{syn}} + x_j(E_j - V_n)R_n^{\text{tot}}$. Update: $\Delta g_j^{\text{syn}} \propto$
548 $\int e_j^{\text{syn}}(t)m_n(t)dt$ [18, 19].

549 G Weight Statistics and Biological References

550 As a secondary analysis, we examine whether the conductance-based update naturally induces charac-
551 teristic weight statistics. Because synaptic updates scale with input resistance $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$, the volt-
552 age equation implies multiplicative weight dynamics, creating a regime analogous to exponentiated-
553 gradient updates. Such multiplicative dynamics suggest log-normal weight distributions in trained
554 networks [38, 39]. We compare the resulting distributions against two biological references: electro-
555 physiological data from Song et al. [38] (rat V1 L5 pyramidal, $\sigma = 0.94$) and the MICrONS mm³
556 connectome [40] (mouse V1 synapse sizes).

557 **Model weight distributions are log-normal.** We fit log-normal distributions to the active (top- K
558 masked) excitatory synaptic weights after training on MNIST and Fashion-MNIST. Table S15 shows
559 the fitted σ (standard deviation of $\log w$) for each condition. Shunting consistently produces narrower
560 distributions than additive integration, and backpropagation produces narrower distributions than
561 local learning. These two factors combine approximately additively: BP+Shunting $\rightarrow$ narrowest,
562 Local+Additive $\rightarrow$ broadest.

563 **Shunting is closest to the biological references considered here.** BP+Shunting produces $\sigma =$
564 0.94 on MNIST, very close to the Song et al. reference ($\Delta = 0.008$). Local+Shunting produces
565 $\sigma = 1.13$ – 1.22 , close to MICrONS E $\rightarrow$ E synapse sizes ($\sigma = 1.14$; $\Delta \leq 0.08$). By contrast,
566 Local+Additive produces $\sigma = 1.43$ – 1.69 , broader than any biological reference considered. This
567 pattern is consistent across datasets (MNIST and Fashion-MNIST), suggesting it reflects structural
568 properties of the learning rule rather than task-specific features.

Condition	MNIST		Fashion-MNIST	
	Exc σ	Inh σ	Exc σ	Inh σ
BP + Shunting	0.94±0.01	0.87±0.01	0.92±0.01	0.84±0.01
BP + Additive	1.30±0.01	1.22±0.01	1.16±0.01	1.05±0.01
Local + Shunting	1.22±0.01	1.71±0.02	1.13±0.06	1.36±0.06
Local + Additive	1.69±0.16	1.57±0.07	1.43±0.16	1.11±0.12
Song et al. 2005	$\sigma = 0.94$ (rat V1 L5, EPSP amplitude)			
MICrONS E→E	$\sigma = 1.14$ (mouse V1, synapse size, $n=3.9M$)			
MICrONS I→E	$\sigma = 0.92$ (mouse V1, synapse size, $n=3.5M$)			

Table S15: **Weight distribution width.** Log-normal σ of excitatory and inhibitory synaptic weights (mean±std across 5 seeds). Biological references are included only as descriptive anchors. With biologically constrained E/I ratios (4:1), shunting self-normalization produces weight heterogeneity in the range spanned by those references, whereas additive integration is broader and less stable.

The match depends on inhibitory conductance. We test sensitivity to the E/I synapse ratio using a sweep over $N_I \in \{0, 2, 5, 10, 20, 30, 40\}$ inhibitory synapses per branch (Appendix Fig. S11). Without inhibition ($N_I=0$), all conditions produce $\sigma > 2.0$, far from biology. With biologically constrained ratios ($N_I=10-20$, corresponding to E/I ratios of 4:1 to 2:1; [42]), shunting networks enter the biological range ($\sigma \approx 0.9-1.2$), while the effect of N_I on additive networks is less systematic. The weight distribution shape is also robust to dendritic depth (branch factors [9] through [3,3,3,3]; see Appendix Fig. S11A): BP+Shunting converges toward Song et al. with increasing depth, while Local+Shunting remains near MICrONS E→E across depths.

H Weight Distribution Sensitivity Analysis

MICrONS connectome analysis. We extracted synapse size distributions from the MICrONS mm³ condensed connectome (v1181; 9.0M connections, 71.9K neurons; 89% excitatory). Splitting by pre/post cell type: E→E connections ($n=3.85M$) show $\sigma_{\text{size}}=1.14$ and CV=1.01; I→E ($n=3.53M$) show $\sigma=0.92$ and CV=1.03; E→I ($n=1.11M$) show $\sigma=0.83$; I→I ($n=0.51M$) show $\sigma=0.85$. Inhibitory connections are more frequently multi-synaptic (I→E: 37.5% multi-synapse; E→E: 11.5%), and inhibitory synapses are on average smaller (median 3532 vs. 4676 for E→E).

Sensitivity to ee_synapses. Using the E/I grid sweep ($N_E \in \{20, 40, 80\} \times N_I \in \{0, 2, 5, 10, 20, 30, 40\}$; 168 models), excitatory σ increases modestly with N_E in both architectures (+0.4–0.6 from $N_E=20$ to 80), but the shunting–additive ordering is preserved at all N_E values. The dominant driver of σ is N_I : going from $N_I=0$ to $N_I=10$ reduces shunting σ by approximately 1.0–1.5 units, while the effect on additive is smaller and less systematic.

Sensitivity to depth. From the depth-scaling sweep ([9], [3,3], [3,3,3], [3,3,3,3]; 80 models, 5 seeds each), BP+Shunting excitatory σ decreases with depth ($1.06 \rightarrow 0.90$), converging toward Song et al. (0.94). Local+Shunting σ increases slightly ($1.18 \rightarrow 1.33$) but remains in the biological range at shallow depths. BP+Additive shows the strongest depth sensitivity ($1.38 \rightarrow 0.79$), while Local+Additive diverges with depth ($2.01 \rightarrow 2.44$). The morphology-aware modulators (path propagation, depth/centrality weighting) have negligible effect on σ .

Mechanistic interpretation. The shunting conductance $g_n^{\text{tot}} = g_n^{\text{leak}} + \sum g_n^{\text{exc}} + \sum g_n^{\text{inh}}$ appears in the update factor $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$. As inhibitory conductance increases, g_n^{tot} grows and R_n^{tot} shrinks, compressing the multiplicative scale of weight updates. This is analogous to a learning rate that self-adjusts via the total conductance: strong synapses contribute more to g_n^{tot} , reducing their own future update magnitude. This acts as a natural form of weight-dependent regularization. The resulting equilibrium distribution is log-normal, consistent with Loewenstein et al. [41] and related multiplicative spine-dynamics analyses. Additive integration lacks this self-normalizing mechanism, producing broader distributions that fall outside the biological range.

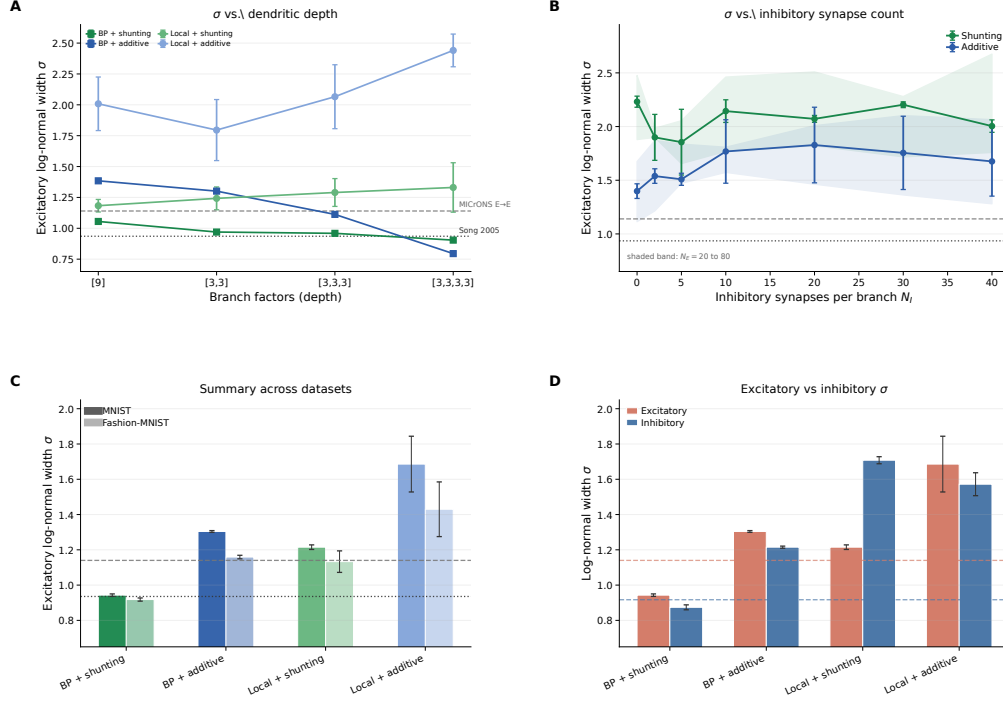


Figure S11: **Synaptic weight distribution sensitivity to architecture.** (A) Log-normal σ vs. dendritic depth (branch factors [9] to [3,3,3,3]). Both backprop and local shunting remain closer to the biological-width references than their additive counterparts across depth. (B) σ vs. inhibitory synapses per branch (N_I). Without inhibition ($N_I=0$), all conditions are overly broad; with biologically constrained $N_I=10$ –20, shunting enters the biological range. Shaded bands show the sweep over $N_E=20$ to 80. (C) Summary of MNIST and Fashion-MNIST results with biological references. The shunting ordering is consistent across datasets. (D) Excitatory vs. inhibitory weight σ on MNIST with MICrONS E→E and I→E reference bands.

I Synthetic Benchmark Descriptions

Each synthetic task is generated procedurally from a fixed random seed and uses fixed train/val/test splits for reproducibility. All tasks use 500-dimensional flattened inputs and 10-class outputs unless otherwise noted.

Context gating. Inputs are Gaussian mixtures over 10 classes, but the class-conditional means are linearly transformed by one of $C = 4$ context configurations. A one-hot context vector (appended to the input) indicates which configuration is active. Without the context, classes are ambiguous; with it, the optimal boundary shifts. This task tests whether credit assignment can exploit the context gating structure to modulate branch-specific representations appropriately. The HSIC auxiliary objective (weight 0.01) is used to decorrelate layer representations, adding roughly 3 pp under local learning.

Noise resilience. Inputs are MNIST features corrupted by structured channel noise: a random 50×50 projection matrix adds correlated noise drawn from $\mathcal{N}(0, \sigma_{\text{task}}^2 I)$ with $\sigma_{\text{task}} = 1.5$. The noise projection is fixed across training; the network must filter structured interference to recover the underlying class signal. Credit assignment must remain coherent under the corrupted signal, making this task particularly sensitive to path-gain dispersion.

Info shunting. Inputs contain two additive streams: a low-frequency signal and a high-frequency distractor drawn with equal energy. A second context input (separate channel) indicates whether the relevant signal is low- or high-frequency on each trial. Correct classification requires the network to gate the distractor stream, a computation that naturally invites inhibitory modulation. Performance

622 at $N_I=0$ is notably below shunting performance (+24.8 pp), reflecting the task’s dependence on
623 divisive normalization.

624 **Cue integration.** Two noisy cue streams (A and B) are presented simultaneously, each carrying
625 partial information about one of 10 classes. A binary context signal indicates which cue is reliable on
626 each trial (cue A or cue B). The fixed-pathway variant duplicates the context into both cue branches;
627 the learned-routing variant requires the router to discover the cue separation from data. This task is
628 used in Sec. 4.7 to test structured pathway-vector broadcast.

629 J Depth Scaling and Noise Robustness

630 **Depth scaling.** Varying dendritic depth from 1–4 layers (branch factors [9] to [3, 3, 3, 3]): shunting
631 local degrades from 63.5% to 57.4%; additive local degrades more steeply from 54.9% to 29.7%.
632 The shunting advantage grows with depth (+8.5 $\rightarrow$ +27.7 pp). Backprop ceilings remain stable at
633 about 90–92%. See Fig. S6A.

634 **Noise robustness.** Gaussian noise $\mathcal{N}(0, \sigma^2)$ on broadcast error: shunting is robust to $\sigma \leq 0.1$ (about
635 62%) while additive drops from 46.5% to chance at $\sigma=1.0$, confirming that shunting credit signals
636 carry useful learning information beyond the broadcast magnitude. See Fig. S6B.

637 NeurIPS Paper Checklist

638 1. Claims

639 Question: Do the main claims made in the abstract and introduction accurately reflect the
640 paper’s contributions and scope?

641 Answer: [Yes]

642 Justification: The abstract and introduction state the paper’s main mechanistic claim, the
643 scope of the conductance-based setting, and the regime-dependent nature of the empirical
644 results; see Sec. 1 and Sec. 6.

645 Guidelines:

- 646 • The answer [N/A] means that the abstract and introduction do not include the claims
647 made in the paper.
- 648 • The abstract and/or introduction should clearly state the claims made, including the
649 contributions made in the paper and important assumptions and limitations. A [No] or
650 [N/A] answer to this question will not be perceived well by the reviewers.
- 651 • The claims made should match theoretical and experimental results, and reflect how
652 much the results can be expected to generalize to other settings.
- 653 • It is fine to include aspirational goals as motivation as long as it is clear that these goals
654 are not attained by the paper.

655 2. Limitations

656 Question: Does the paper discuss the limitations of the work performed by the authors?

657 Answer: [Yes]

658 Justification: The discussion explicitly narrows the scope of the claims and discusses
659 limitations involving benchmark strength, broadcast assumptions, depth scaling, and the
660 exploratory stress tests; see Sec. 6.

661 Guidelines:

- 662 • The answer [N/A] means that the paper has no limitation while the answer [No] means
663 that the paper has limitations, but those are not discussed in the paper.
- 664 • The authors are encouraged to create a separate “Limitations” section in their paper.
- 665 • The paper should point out any strong assumptions and how robust the results are to
666 violations of these assumptions (e.g., independence assumptions, noiseless settings,
667 model well-specification, asymptotic approximations only holding locally). The authors
668 should reflect on how these assumptions might be violated in practice and what the
669 implications would be.

- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: The voltage model, assumptions, theorem statements, and proof details are given in Sec. 2 and Appendix C; proof sketches and the exact factorization sanity check are cross-referenced in the main text.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The main paper and appendices specify the datasets, architectures, local-rule family, broadcast modes, representative hyperparameters, and compute regime needed to reproduce the headline results; see Sec. 4 and Appendix B.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.

- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: To preserve double-blind review, the anonymous submission does not include a public repository or anonymized code bundle. The paper instead provides architectural, algorithmic, and configuration details and states that code and configs will be released after review; see Appendix B.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: The main text states the benchmark families, seed protocol, representative local-rule settings, and major training choices, while the appendices provide implementation details and operating-mode inventories; see Sec. 4 and Appendix B.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: The paper reports means and standard deviations across 3–5 seeds for headline results, states this explicitly in the text, and shows error bars or seed variability in the relevant tables and figures; see Sec. 4 and Appendix A.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix B describes the internal GPU cluster setting, the use of single A100/H100-class GPUs, rough per-run runtime scales, and the fact that broader exploratory sweeps required additional compute beyond the final reported runs.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.

- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The work uses standard public benchmarks and mechanistic simulations, does not involve human-subjects experimentation or restricted data collection, and we are not aware of any aspect that violates the NeurIPS Code of Ethics.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The discussion includes a broader-relevance and potential-harms paragraph that distinguishes mechanistic neuroscience relevance from possible misuse through overclaiming biological plausibility or overstating robustness; see Sec. 6.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The submission does not release a high-risk generative model, foundation model, or scraped dataset, so the safeguard question is not applicable here.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [No]

Justification: The paper credits all benchmark and connectomic assets and states that they are used under their published access conditions, but it does not enumerate license names or terms of use explicitly for each asset in the anonymous submission; see Appendix B.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [N/A]

Justification: The paper does not release a new dataset, benchmark, or pretrained model asset in this anonymous submission.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.

- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The work does not involve crowdsourcing or direct research with human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The work does not involve crowdsourcing or human-subjects research requiring IRB review.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: LLMs are not part of the proposed method, theorem, experiments, or analysis pipeline, so no declaration is required for the core scientific contribution.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.